

ARISTOTLE UNIVERSITY OF THESSALONIKI
FACULTY OF ENGINEERING
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING



Accelerating top-k selection on heterogeneous CPU, GPU and NPU systems

DIPLOMA THESIS

Ioannis Michalainas

ioannisam@ece.auth.gr

Supervisor: Nikolaos Pitsianis

Associate Professor, School of Electrical and Computer Engineering

Thessaloniki, August 2026

Abstract

Note on this translation

Warning! This document was translated from Greek into English automatically with the help of a large language model. The original Greek text is the authoritative version of this thesis. Some passages in this translation may read awkwardly or imprecisely as a result of the automated process.

Top-k selection, the task of identifying the k largest elements of a set, appears across a wide range of applications and is repeated often enough that its cost is noticeable. The recent integration of neural processing units into the same package as the processor makes the question of whether offloading such computations to an accelerator is worthwhile a practical one.

This work implements two algorithms with fundamentally different behaviour, truncated bitonic sorting which is data-oblivious and map-reduce selection with local heaps which is data-dependent, on three architectures: a processor with AVX-512 vector extensions, a GPU programmed in CUDA, and an AMD XDNA neural processing unit. The last was built entirely at the level of the MLIR-AIE framework, as no top-k selection library exists for that architecture. All six implementations were measured over the same parameter space, under two explicitly distinct measurement scopes, recording time, energy and the volume of data moved and placing every run against a roofline measured on the same machine.

The results show that the problem is limited by data movement on every architecture, that the ranking of the two algorithms inverts with the ratio n/k rather than with the architecture and that the choice of measurement boundary completely reverses the ranking of the architectures. In energy terms the neural processing unit exhibits the lowest power draw and simultaneously the highest consumption, since duration dominates the product. No accelerator run is Pareto-optimal: for small k , offloading is measurably worse than computing on the processor.

Keywords: top-k selection, bitonic sorting networks, heterogeneous computing, neural processing unit, MLIR-AIE, SIMD vectorisation, energy efficiency, roofline model

Acknowledgements

This thesis is the largest programming project I have undertaken to date. I would like to thank my supervisor, Prof. Nikolaos Pitsianis, in whose courses I discovered my interest in high performance computing and low-level programming.

I would also like to thank my family for their unwavering support throughout my studies, especially my brother Manolis, who, as a fellow engineer, helped me with valuable advice and guidance.

Ioannis Michalainas

Contents

Abstract	2
Acknowledgements	3
1 Introduction	9
1.1 Scope of the Thesis	9
1.2 Motivation	9
1.3 Contribution	10
1.4 Structure of the Thesis	11
2 Background and Related Work	12
2.1 The top-k selection problem	12
2.2 Selection algorithms	13
2.2.1 Bitonic sorting networks	14
2.2.2 Truncated bitonic sort	14
2.2.3 Selection with heaps and map-reduce	16
2.2.4 Radix selection	17
2.3 Processing architectures	17
2.3.1 CPU vector extensions	17
2.3.2 The SIMT model and GPU memory hierarchy	18
2.3.3 The AIE dataflow architecture	18
2.4 Existing implementations and libraries	19
3 Methodology and experimental setup	21
3.1 Reference system	21
3.2 Structure of the measurement harness	22
3.3 Measured quantities	23
3.3.1 Measurement scopes: algo and end-to-end	23
3.3.2 Timing	24
3.3.3 Energy measurement	25
3.3.4 Data traffic measurement	26
3.4 Roofline model	27
3.5 Baseline libraries	29
3.6 Correctness checking	30
3.7 Parameter space	30
3.7.1 Data types	30
3.7.2 Input distributions	31
3.8 Measurement conditions and reproducibility	32

4	CPU Implementation	34
4.1	Design decisions	34
4.1.1	Vectorisation	34
4.1.2	Vector type abstraction	35
4.1.3	Half-precision floating-point support	36
4.1.4	Parallelisation across multiple threads	36
4.2	Bitonic sort	37
4.3	Map-reduce with local heaps	39
4.4	Memory management and data traffic	40
4.5	Reference implementation	41
4.6	Limiting factors	41
5	Implementation on the GPU	43
5.1	Design choices	43
5.1.1	Mapping onto the grid and thread blocks	43
5.2	Bitonic sort	44
5.2.1	Organisation into tiles and shared memory	44
5.3	Map-reduce with local heaps	46
5.3.1	Parallelism limit for large k	47
5.4	Memory management and data traffic	48
5.5	Ground truth	49
5.5.1	Radix selection in CUB	49
5.6	Bottlenecks	50
6	Implementation on the NPU	52
6.1	Design choices	52
6.1.1	Development flow with MLIR-AIE	52
6.1.2	Tile topology and object_fifo	53
6.2	Bitonic sort	53
6.2.1	Compute tile kernel	54
6.2.2	Scaling to multiple tiles	54
6.3	Map-reduce with local heaps	55
6.4	Memory management and data traffic	55
6.4.1	Pipelining and the role of the host	55
6.4.2	Data encoding and types	56
6.4.3	Final correction on the host	57
6.5	Ground truth implementation	57
6.6	Limiting factors	57
7	Results and evaluation	59
7.1	Correctness	59
7.2	Execution time	60
7.2.1	Dependence on input size	60
7.2.2	Dependence on the parameter k	62
7.3	Comparison across architectures	66
7.4	Roofline analysis	68
7.4.1	Memory bandwidth utilisation	70
7.5	Energy assessment	72
7.5.1	Energy-delay product	75

7.6	Effect of data type	76
7.7	Sensitivity to the input distribution	77
8	Conclusions and Future Work	80
8.1	Conclusions	80
8.2	Limitations	81
8.3	Future Work	82
A	Acronyms	84
B	Reproducing the Measurements	85
B.1	Software Requirements	85
B.2	Compilation	86
B.3	Running the Experiments	86
	B.3.1 Running Individual Steps	87
	B.3.2 Execution Parameters	87
B.4	Generating the Plots	88

List of Figures

2.1	Selection versus sorting	13
2.2	The full bitonic network and its truncation	15
2.3	Heap-based selection and the map-reduce scheme	16
2.4	Radix selection	17
3.1	The four levels of the measurement harness	23
3.2	The two measurement scopes per architecture	24
3.3	Reading the roofline model	28
3.4	The five input distributions	31
4.1	The fundamental compare-exchange motif	35
4.2	The half-precision path	36
4.3	Fusing consecutive layers into one pass	38
4.4	Bulk rejection in map-reduce	39
5.1	Sorted block and shared-memory tile	46
5.2	The parallelism limit of map-reduce	48
6.1	Tile topology and object_fifo queues	53
6.2	The three-stage pipelining of the offload	56
7.1	Pass rate of the correctness check for the two algorithms of this work, by architecture.	59
7.2	Time as a function of size, under the two measurement scopes	60
7.3	The same algorithm across the three architectures, for k equal to 8.	61
7.4	Time as a function of size, by value of k	63
7.5	The three algorithms on the GPU for k equal to 131072.	65
7.6	Speedup relative to the ground-truth implementation	66
7.7	Placement of the runs against the measured ceilings	68
7.8	Measured read bandwidth as a function of working-set size, for the three architectures.	69
7.9	Utilisation of the two ceilings on the CPU, for k equal to 8.	70
7.10	Achieved bandwidth by architecture	71
7.11	Algorithmic and end-to-end time on the GPU	72
7.12	Net energy by architecture and algorithm, for k equal to 8.	73
7.13	Net power by architecture and algorithm, for k equal to 8.	73
7.14	End-to-end time and energy per measured case	74
7.15	Energy-delay product as a function of input size.	75
7.16	Sensitivity of time to the input distribution	78

List of Tables

3.1	The reference system on which all measurements were carried out. . .	21
3.2	The content of the two measurement scopes per architecture.	23
3.3	The data types that are swept.	31
3.4	The input distributions that are swept.	31
3.5	The conditions to which the system is pinned before each sweep. . . .	32
6.1	Time breakdown of the offload region by stage.	56
7.1	Algorithmic time for the two extremes of k	64
7.2	Algorithmic time by data type	76
7.3	Algorithmic time by input distribution	77
B.1	The execution parameters and their defaults.	88

Chapter 1

Introduction

1.1 Scope of the Thesis

This thesis examines the acceleration of top-k selection, i.e., identifying the k largest elements of a set of n values, on three different architectures: a general-purpose processor, a graphics processing unit, and a neural processing unit integrated into the same package as the processor.

For each architecture, two algorithms with fundamentally different behaviour are implemented: truncated bitonic sort, which executes a fixed sequence of comparisons independent of the data, and map-reduce selection with local heaps, which adapts its work to the input values. All six implementations are measured over the same parameter space, with a common harness, common correctness checking, and common measurement semantics, so that the differences that emerge can be attributed to the algorithms and the hardware rather than to the way they are measured.

The question posed is not simply which architecture is faster. It is whether, and under what conditions, offloading the computation to an accelerator is worthwhile, once the full cost of the offload is accounted for and not just the kernel time. As will be shown, the two formulations of the question give opposite answers.

1.2 Motivation

Neural processing units have only recently ceased to be data-centre equipment. They are now integrated on the same die as processors in consumer laptops, share the same main memory, and are available to any application that could make use of them. The question "is it worth offloading this computation to the unit?" has thus shifted from infrastructure design to the programmer's everyday practice.

The answer is not obvious, because these units were designed for a specific purpose. Their software ecosystem is organised around neural network operators, and their architecture rests on the assumption that every piece of data transferred will be used many times over. What happens when the workload does not have this shape remains largely unexplored.

Top-k selection is a suitable vehicle for posing this question. It is a problem simple to

state but widespread in practice, appearing in nearest-neighbour search, in retrieval and recommendation systems, and in the sampling of language models. At the same time, it sits at the opposite end of the spectrum from the workloads accelerators were designed for: it performs minimal operations per element and reuses almost nothing. If offloading pays off even here, it will pay off almost everywhere; if it does not, we have a clear indication of the limits of usefulness of these units.

A second, independent motivation is methodological. Accelerator comparisons usually report kernel execution time, concealing the cost of transferring data to and from the device. Yet this cost is exactly what determines whether offloading is worthwhile. This thesis systematically measures both, and shows that the choice of measurement boundary does not merely change the numbers, but completely reverses the ranking of the architectures.

1.3 Contribution

The contribution of this thesis is fourfold.

Implementations. Six implementations were built, two algorithms across three architectures, with support for five data types. The processor implementation makes use of AVX-512 vector extensions and multiple threads, the graphics processing unit implementation uses an organisation into tiles with shared memory, while the neural processing unit implementation was built entirely at the level of the MLIR-AIE framework, as no top-k selection library exists for that architecture. The correctness of every combination was verified on 2700 cases, all successful.

Measurement harness. A harness was developed that measures every case over two explicitly distinct scopes, the algorithmic and the end-to-end. It records energy with common semantics across all three architectures, tallies the volume of data moved, and places every run against a roofline measured on the same machine. The harness is independent of the specific problem and applicable to any workload being considered for offloading to an accelerator.

Findings. The experimental evaluation yields four conclusions. Top-k selection is a data-traffic problem on every architecture examined, and no implementation is bound by computational throughput. The ranking of the two algorithms inverts with the ratio n/k rather than with the architecture, with a difference reaching three orders of magnitude. The choice of measurement boundary completely reverses the ranking of the architectures. Finally, low power draw does not imply low energy consumption: the neural processing unit has the lowest power of all implementations and at the same time consumes several times more energy (because it takes several times longer).

Reproducibility. The entire experimental procedure, from building the executables to producing the plots, is carried out by scripts included in the repository. The measurement conditions are explicitly stabilised and recorded together with the results. Details are given in Appendix B.

1.4 Structure of the Thesis

Chapter 2 defines the problem, presents the families of algorithms that solve it, and describes the execution models of the three architectures, concluding with the existing implementations and the gap this thesis addresses.

Chapter 3 describes the reference system and the experimental setup: exactly what is measured, under what semantics, with what comparison metrics, and under what conditions. This chapter is a prerequisite for reading the results.

Chapters 4, 5 and 6 present the implementations on each of the three architectures. They follow a common structure, so that the corresponding sections can be compared directly: design choices, the two algorithms, memory management and data traffic, ground truth, and finally, the limiting factors identified.

Chapter 7 presents and interprets the results: correctness, execution time, comparison of architectures, analysis with the roofline model, energy evaluation, and finally, the effect of data type and input distribution.

Chapter 8 summarises what generalises, states the limits within which it holds, and proposes directions for future work.

The appendices include a list of acronyms and instructions for reproducing the measurements.

Chapter 2

Background and Related Work

This chapter defines the problem, presents the algorithm families that solve it, and describes the execution models of the three architectures. The goal is not exhaustive coverage, but laying the groundwork needed to read chapters 4 through 6. The specific implementations of this work are presented there, not here.

2.1 The top-k selection problem

Given an array A of n elements from a totally ordered set and an integer $k \leq n$, the top-k selection problem asks for the subset of the k largest elements of A . This work considers the form that asks only for this set, not the order of the elements within it, since this is the form required by applications and the only one against which algorithms with different internal organisation can be fairly compared.

The critical observation is that the problem asks for strictly less information than sorting. Full sorting determines the relative order of all n elements, whereas selection is satisfied with separating them into two groups. This difference is captured by the lower bounds: every comparison-based algorithm requires $\Omega(n \log n)$ comparisons to sort, but only $\Theta(n)$ to select with fixed k [1]. Selection is therefore substantially cheaper, and any implementation that sorts the entire input in order to keep k elements performs more work than the problem requires.

Figure 2.1 illustrates this difference.

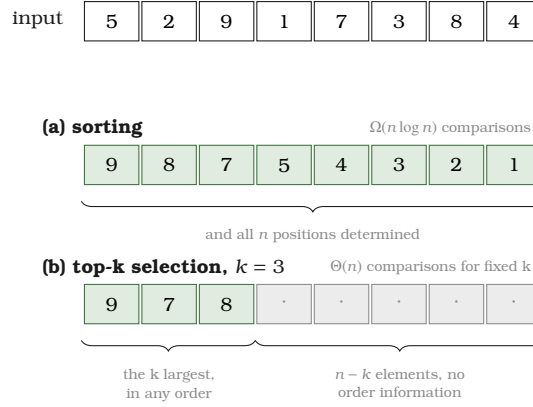


Figure 2.1: The same input under the two requirements. Sorting (a) determines all n positions, while selection (b) splits the elements into only two groups: the k largest, in any order, and the rest, for which no order information is kept.

This gap between sorting and selection is not constant; it depends on the ratio k/n . At one extreme, when k is much smaller than n , selection is nearly linear and sorting wastes almost all of its work. At the other, when k approaches n , the two problems nearly coincide and the margin disappears. The entire work moves between these two extremes. Chapter 7 shows that the ranking of the algorithms inverts as this gap is traversed.

The problem appears across a multitude of applications. K-nearest-neighbour search requires selecting the smallest distances from a distance matrix [2]. Information retrieval and recommendation systems return the k best results of a query. In language models, top- k sampling and beam search perform selection at every generation step. A common feature of these cases is that selection is not the primary workload but is repeated often, so its cost is noticeable precisely because it is added at every iteration.

2.2 Selection algorithms

Selection algorithms fall into families based on a criterion that proves decisive for acceleration: whether the sequence of operations depends on the data values or not. An algorithm whose schedule is fixed executes the same operations in the same order regardless of the input. A data-dependent algorithm adapts its course, performing less work on favourable inputs and more on adverse ones.

This distinction, which corresponds to the established classification of parallel selection algorithms into sorting, partial sorting, and partitioning methods [3], is not only a matter of theoretical complexity. It determines which hardware can execute the algorithm efficiently, because parallel architectures presuppose that many processing elements execute the same instruction. It also determines whether execution time is predictable, a question examined experimentally in section 7.7.

2.2.1 Bitonic sorting networks

A sorting network is a fixed sequence of comparators. Each comparator takes two array positions, compares their contents, and swaps them if they are in the wrong order. The positions it compares are fixed in advance and do not depend on the values. The network is therefore the standard example of a data-oblivious algorithm: the same sequence of comparisons is always executed, and the only thing that changes is the outcome of each comparison.

The comparators are organised into *stages*. Within a stage, no position appears in two comparators, so all comparators of a stage can execute simultaneously. The stages, however, must execute sequentially. The number of stages thus constitutes the depth of the network and defines the lower bound on time on unboundedly parallel hardware.

Batcher's construction rests on the notion of a bitonic sequence [4]. A sequence is called bitonic if it first increases and then decreases, or if it results from such a sequence by a cyclic shift. The fundamental property is that a bitonic sequence of length $2m$ can be sorted in $\log_2(2m)$ stages: comparing each element with the one m positions away yields two bitonic sequences of length m , such that every element of the first is less than or equal to every element of the second. The two halves are then sorted independently by the same procedure.

Full sorting of n elements results from repeated application: pairs of sorted blocks of opposite direction form a bitonic sequence of double the length, which is sorted to yield a block of double the size. The network has $\log_2 n (\log_2 n + 1)/2$ stages and $n/2$ comparators per stage, i.e., $O(n \log^2 n)$ comparators in total.

This count is larger by a factor of $\log n$ than the $O(n \log n)$ bound achieved by sequential sorting algorithms. Bitonic sort thus performs more comparisons than necessary. The trade-off is that these comparisons are entirely predetermined and uniform, a property that makes them executable on vector units, on SIMD models, and on dataflow arrays without any control-flow divergence. This trade-off, more operations in exchange for perfect regularity, is why sorting networks remain competitive on parallel hardware despite their asymptotic disadvantage.

2.2.2 Truncated bitonic sort

Full sorting produces far more information than selection requires. Truncated bitonic sort exploits this fact by discarding elements as soon as it is proven that they cannot belong to the result [2]. The same idea, halving the active array at every step, also underlies partial-sorting implementations for GPUs [5].

Figure 2.2 contrasts the two structures. Let k' be the smallest power of two not less than k . The procedure has two phases. In the first, ordinary bitonic sort is executed, but it stops as soon as sorted blocks of size k' are formed, instead of continuing to the single block of size n . In the second, blocks are merged in pairs; of the $2k'$ elements in each pair, only the k' largest are kept and the rest are discarded permanently. This step halves the size of the active array and repeats until a single block remains, which contains the desired result.

The correctness of this discarding is immediate. If an element is not among the k'

largest of two sorted blocks, then there are already k' elements larger than it in the array, so it cannot be among the k largest of the whole set.

The gain is twofold. On the one hand, the first phase requires $\log_2 k' (\log_2 k' + 1)/2$ stages instead of $\log_2 n (\log_2 n + 1)/2$, a count that depends on k rather than on n . On the other hand, the stages of the second phase act on an array that shrinks geometrically, so their cost sums to a term linear in n . The overall complexity falls from $O(n \log^2 n)$ to $O(n \log^2 k)$.

For $n = 2^{24}$ and k equal to 8, the full network has 300 stages and 2 516 582 400 comparators, whereas the truncated network has 90 stages and 92 274 668 comparators, i.e., twenty-seven times fewer. For k equal to 131072, however, the figures are 279 stages and 1 441 595 392 comparators, only 1.7 times fewer than the full network. The benefit of truncation evaporates as k grows, a fact that partly explains the behaviour recorded in section 7.2.2.

Crucially, truncation does not sacrifice data-obliviousness. Which elements are discarded depends on the values, but *how many* and at which steps does not: the schedule of stages is computed in advance from n and k alone. This property preserves suitability for parallel hardware and is confirmed experimentally in section 7.7, where the implementation exhibits a variation of under twelve percent across five very different input distributions.

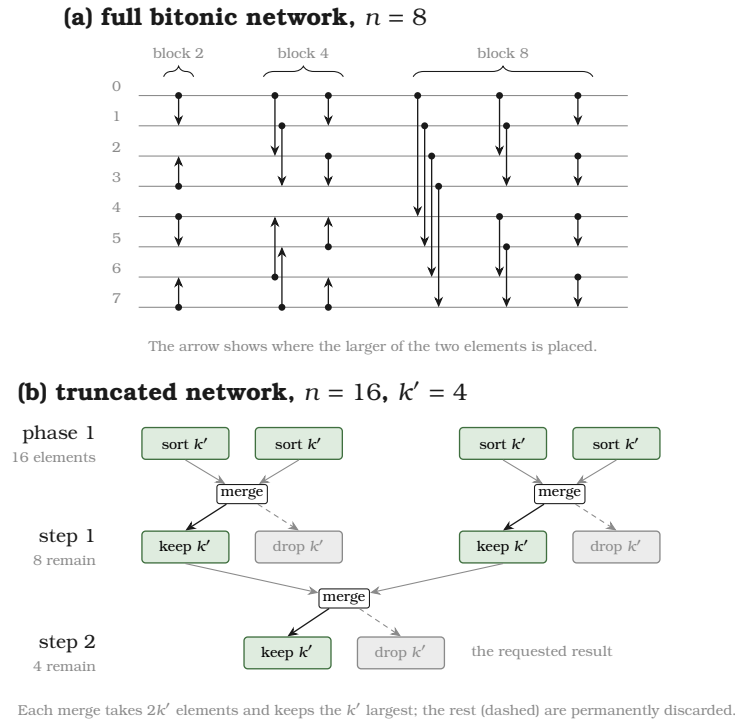


Figure 2.2: The full bitonic network and its truncation. In (a) each horizontal line is an array position and each vertical arrow a compare-exchange. The network is data-independent, since it always executes the same sequence of comparisons in the same order, regardless of the values. In (b), when only the k' best elements are requested, the first phase stops as soon as sorted blocks of size k' have formed, and in the second phase each pairwise merge keeps only the k' best of the $2k'$ compared. The array size is halved at each step, so subsequent steps cost progressively less.

2.2.3 Selection with heaps and map-reduce

The second family follows the opposite logic. Instead of imposing a fixed schedule, it maintains a structure holding the k best elements examined so far, and updates the structure only when necessary.

The structure is a min-heap of capacity k . Its root holds the smallest of the k current best elements and acts as an admission threshold. For each new element, a single comparison against the root is performed. If the element does not exceed it, it is discarded with no further work. Otherwise it replaces the root, and the heap is restored in $O(\log k)$ steps.

The asymmetry between the two cases determines the algorithm's behaviour. On random input, the threshold rises quickly and the overwhelming majority of elements are discarded with a single comparison, so the total time approaches linear. On ascending input, however, every new element exceeds the root and forces a restoration, giving $O(n \log k)$ in the worst case. The same code thus performs a fundamentally different volume of work depending on the order in which values appear, and section 7.7 measures this difference at a factor of seventy-two.

Parallelisation follows the map-reduce scheme, as shown in figure 2.3. In the map phase, the input is partitioned and each worker maintains its own local heap, without synchronising with the others. In the reduce phase, the partial results are merged into one. This scheme is attractive because the map phase requires no communication, but it imposes a limit that proves critical: each local heap must first fill with k elements before it can start filtering, so the number of workers that actually contribute cannot exceed the ratio n/k . This limit is negligible on a processor with eight threads, but disastrous on a GPU with thousands, as analysed in section 5.3.1.

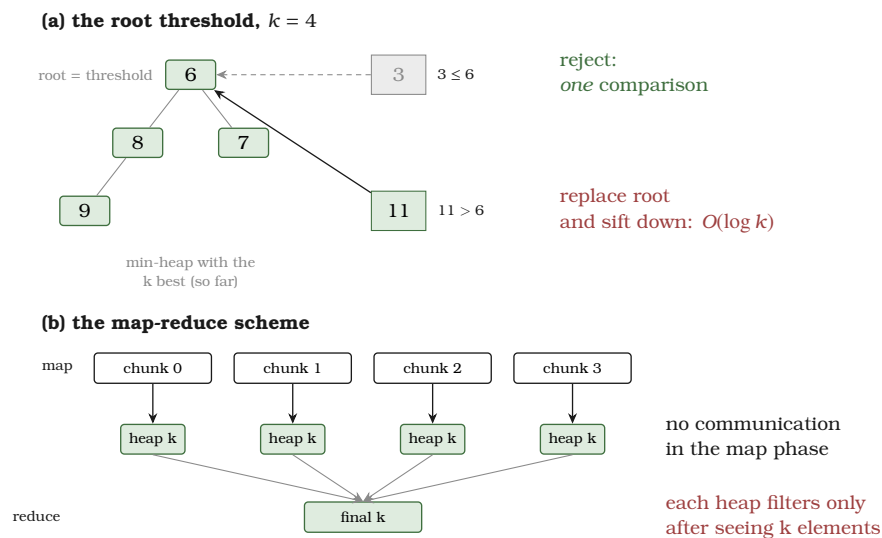


Figure 2.3: Heap-based selection and its parallelisation. In (a) the root of the min-heap acts as a threshold: an element that does not exceed it is rejected with one comparison, while one that exceeds it replaces the root and triggers a sift-down. In (b) the input is partitioned into chunks, one heap per worker, and the partial results are merged into one.

2.2.4 Radix selection

The third family does not compare elements against each other but examines their binary representation. Values are distributed into buckets according to a group of bits, the bucket counts are summed, and from these it follows which bucket contains the k -th largest element. Buckets above it belong entirely to the result, buckets below it are discarded entirely, and the search continues only within the critical bucket, using the next group of bits.

The procedure is shown in figure 2.4. Two properties stand out. First, the number of passes is determined by the width of the data type and not by k , so the method exhibits no collapse point as k grows. Second, each pass is a sequential read with frequency counting, a pattern particularly favourable for high-bandwidth architectures.

This method is also data-dependent, but in a way different from heaps: what matters is not the order in which values appear but the distribution of their bits. When many elements share the same significant bits, the critical bucket narrows slowly and more passes are required; this case is known in the literature as a radix-adversarial distribution and is addressed with adaptive candidate-storage strategies [3]. This family was not implemented in the present work, but it underlies the GPU’s reference library and is examined in more detail in section 5.5.1.

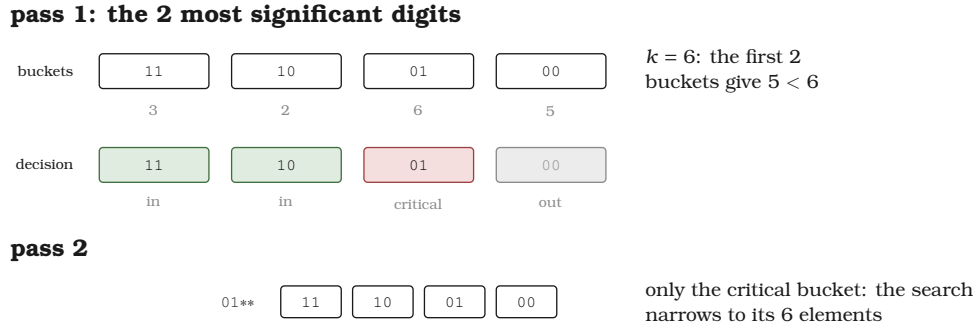


Figure 2.4: Radix selection. At each pass the elements are distributed into buckets according to a group of digits and the count of each is tallied; from the cumulative counts it follows which buckets belong entirely to the result, which are rejected, and which is the *critical* one. No comparison between values is ever performed.

2.3 Processing architectures

The three architectures examined by this work expose parallelism in fundamentally different ways. This section describes their execution models to the extent required to understand the design choices of chapters 4 through 6.

2.3.1 CPU vector extensions

The modern general-purpose processor combines two forms of parallelism. The first is multiple threads, each executing an independent instruction stream. The second is vector extensions, which apply the same operation to multiple data items simultaneously within a single thread.

An AVX-512 register is 512 bits wide and holds sixteen 32-bit integers. Beyond arithmetic operations, its repertoire includes shuffle instructions that permute elements within the register and min/max instructions that select element-wise without branching. The combination of shuffle, min, and max implements exactly one sorting-network comparator, a fact that makes bitonic sort a natural candidate for vectorisation.

The limiting factor is usually not the rate of operations but their supply of data. The cache hierarchy partly compensates for the speed gap with main memory, but only for algorithms that reuse data while it remains at a nearby level. An algorithm that performs many sequential passes over the entire array, such as a naive translation of a sorting network into code, benefits from this not at all. Addressing this problem is the central theme of section 4.2.

2.3.2 The SIMT model and GPU memory hierarchy

The GPU is organised around the SIMT model. Threads are grouped into warps that execute the same instruction on different data. When the threads of a warp follow different branches, the branches execute serially with the inactive threads disabled. Control-flow divergence therefore has a direct cost, and branch-free algorithms have an advantage.

Warps are grouped into thread blocks, and blocks into a grid. Each block has shared memory, a small region managed explicitly by software, with latency far lower than global memory. Shared memory is the tool by which a multi-pass algorithm is converted into a single-pass one: if the chunk of data processed by a block fits there, all intermediate steps execute without returning to global memory.

Access to global memory is efficient only when the threads of a warp request consecutive addresses, so that the individual requests are coalesced into few transactions. Finally, the GPU is a discrete device: data must be transferred to and from its memory over a bus whose bandwidth is an order of magnitude lower than that of local memory. This transfer is the reason this work explicitly distinguishes two measurement scopes, as defined in section 3.3.1.

2.3.3 The AIE dataflow architecture

The term neural processing unit first appeared with a meaning different from today's. Its first registered use referred to an array of micro-machines dynamically configured as neurons based on stored connection information [6], while its first use in the literature described a specialised unit that replaces portions of general-purpose code with approximate neural computation, aimed at saving energy rather than accelerating machine learning [7]. The modern meaning was shaped by the first accelerators dedicated to neural networks [8] and was established with the analysis of the first production unit deployed in a data-centre setting, which documented that specialised architectures outperform processors and GPUs in performance per watt [9].

Central to understanding these units is the design principle stated explicitly in the Eyeriss architecture: in these workloads, moving a piece of data costs orders of

magnitude more energy than the operation to be performed on it, so the design must be organised around maximising local data reuse [10]. This principle is fully vindicated in convolution and matrix multiplication, where every element loaded participates in many operations. The top-k selection problem, however, lies at the exact opposite extreme: each element is examined a minimal number of times and there is essentially no reuse to exploit. This observation foreshadows the central finding of chapter 7 and recurs in section 7.5.

The architecture examined in this work belongs to AMD’s AI Engine family, which was introduced with the Versal adaptive systems [11] and subsequently integrated, as XDNA, into general-purpose processors [12]. This integration is what makes the question posed by this work practical: the unit is no longer a data-centre component but resides on the same die as a laptop’s processor, shares the same main memory, and is available to any application that could make use of it.

The execution model is spatial, fundamentally different from the previous two. The unit consists of an array of tiles, each with its own vector processor and its own local memory. There is neither shared memory nor cache memory: communication happens exclusively through explicit data transfer over an interconnect network.

The consequence is that the program is not expressed as a sequence of operations on a shared address space, but as a graph of compute kernels connected by streaming channels. The programmer specifies which tiles execute which kernel and how data flows between them. The abstraction used in this work for this purpose is the `object_fifo` of the MLIR-AIE framework [13], which is presented in section 6.1.2.

Two properties of the model have direct consequences for top-k selection. The first is that the local memory size of each tile is small and fixed, so the computation must be broken into chunks that fit within it, with the partial results merged elsewhere. The second is that feeding the array is the host’s responsibility, and it must prepare and forward every chunk. When the computation per element is small, as is the case in selection, the cost of feeding dominates over the cost of computation; section 6.4.1 confirms this with a direct measurement.

2.4 Existing implementations and libraries

The availability of ready-made implementations differs dramatically across the three architectures, and this disparity shapes both the question posed by this work and the choice of baselines in section 3.5.

For the processor, the C++ standard library offers `std::nth_element`, which partitions the array around the k-th element in expected linear time, and `std::partial_sort`, which additionally sorts the result by maintaining a heap of size k. Both are universally available and well-tested, but strictly single-threaded: they exploit neither multiple threads nor the hardware’s vector extensions.

For the GPU, CUB, part of the CCCL library collection [14], provides `DeviceTopK`, a radix selection implementation as described in section 2.2.4. It is a mature and highly optimised solution, which this work uses as a baseline without attempting to surpass it across the entire parameter range.

It is worth noting that the extent to which algorithm choice depends on k is also recognised in industry. Recent work on AMD GPUs identifies exactly the same inversion recorded in section 7.2.2 and addresses it with an adaptive strategy: register-level bitonic sort for small k and radix selection for large k , with an empirically determined transition point [15]. The fact that an independent implementation on different hardware arrives at the same distinction reinforces the conclusion that the inversion is a property of the algorithms rather than of the specific platform.

Alongside exact methods, a distinct line of *approximate* algorithms has also developed, which trade some recall for speed and target primarily machine-learning workloads, where top- k selection proves to be a systematic bottleneck on accelerators designed for dense matrix multiplication [16]. The present work is restricted to exact selection, as required by the correctness check of section 3.6.

For the neural processing unit, no equivalent library exists. Its software ecosystem is oriented towards neural-network operators and does not include a top- k selection primitive. The closest available material is an official bitonic sort example on a single AI Engine tile [17], which, however, performs *full* sorting, only for single-precision numbers, and for sizes up to 1024 elements that fit in one tile’s local memory. The gap from the problem addressed by the present work, namely selection rather than sorting, five data types, and inputs of up to 2^{24} elements requiring pipelining across multiple tiles, precisely defines the void this work is called to fill.

The absence of a ready-made solution has two consequences. First, every implementation must be built from scratch at the level of MLIR-AIE [13], a fact that makes this unit the most demanding part of the work. Second, there is no natural on-device baseline, so this work uses the host implementation as ground truth, a choice justified in section 6.5.

Overall, the literature on top- k selection for GPUs is extensive and covers both sorting-network-based and radix-based methods [2, 15]. By contrast, the corresponding literature for AIE-type dataflow accelerators is limited, as these units are recent and their use focuses on convolution and matrix-multiplication workloads. This gap is precisely where the present work positions itself.

Chapter 3

Methodology and experimental setup

3.1 Reference system

All measurements in this work come from a single system, which hosts all three architectures being compared: a general-purpose processor (CPU), a discrete graphics unit (GPU) and a neural processing unit (NPU) integrated into the same package as the processor. This choice isolates differences between implementations from differences between platforms, but introduces a constraint that shapes the measurement conditions: the system is a laptop, and the three units share a common thermal and power budget. Handling of this constraint is described in section 3.8.

System	Lenovo Legion 5 15AHP10
Operating system	Arch Linux x86_64, kernel 7.1.5-arch1-2
Main memory	32 GiB DDR5-5600, dual channel
CPU	AMD Ryzen 7 260
Cores / threads	8 / 16, up to 5.10 GHz
Cache	8 MiB L2, 16 MiB L3
Vector extensions	AVX-512
GPU	NVIDIA GeForce RTX 5060 Laptop
Memory	8 GiB
Software	driver 610.43.03, CUDA 13.3
Power limits	50 W default, 115 W maximum
NPU	AMD XDNA (RyzenAI-npu1)
Software	XRT 2.21.75, firmware 1.5.5.391

Table 3.1: The reference system on which all measurements were carried out.

The processor has eight physical cores with simultaneous multithreading. All measurements run with eight threads, one per physical core, so that no two threads share the same vector unit. The cache hierarchy determines the knee points of the measured bandwidth scale (section 3.4): depending on the input size and the data type, the working set may fit in L2, in L3, or only in main memory, and each im-

plementation’s performance is compared against the ceiling that corresponds to its own working set.

Main memory is arranged in two channels, with two identical 16GiB modules at 5600 MT/s. This detail matters, since a single-channel arrangement would halve the available bandwidth, so any implementation limited by memory would give substantially different results on an otherwise identical system.

The neural processing unit has no memory of its own; it is fed from the same DDR5 through the same memory controllers as the processor, on the same silicon package. This has two consequences that recur throughout this work: the cost of preparing data on the host is not separable from the cost of the computation itself, and the energy meter used covers the whole package rather than the AIE tiles individually (section 3.3.3).

The figures in table 3.1 were verified by the system itself rather than taken from manufacturer datasheets. Characteristics that are not directly measurable, such as the microarchitecture’s marketing name, theoretical bandwidth, or advertised TOPS figures, are not used anywhere in the comparison. Every bandwidth and compute ceiling against which the implementations are compared is measured on the same machine, by the procedure of section 3.4.

3.2 Structure of the measurement harness

The measurement harness is organised into four layers, with one principle running through the design: the measurement procedure is common to all architectures, and only the compute kernel differs. No implementation controls how it is itself measured, so that none can, even unintentionally, time something different from the others.

The first layer is a shared C++ core used by all three implementations. It is responsible for reading parameters, generating the input, scheduling the stages of the bitonic network, reading the energy meters, the warm-up and measurement iteration loop, correctness checking, and printing the report. Because this loop exists in a single place, the timing and energy measurements of all architectures are defined by the same code.

The second layer consists of the per-architecture executables. Each implements the same interface for the three algorithms compared, namely bitonic sort, map-reduce with local heaps, and the ground truth, providing only the function that performs the computation. This function is called by the shared measurement loop of the first layer.

The third layer is the sweep driver. It enumerates the parameter space of section 3.7 and runs each case as a separate process, recording its output verbatim along with its outcome. Running in separate processes ensures that each case starts from a clean memory state and that the failure of one does not invalidate the sweep.

The fourth layer converts the raw results into tables and plots. The analysis operates on the same report text that the user reads: there is no separate output path to the tools, so the values shown in the plots are by construction the values printed by the

executable. The execution environment is stored alongside the results, as described in section 3.8. The commands that produce each stage are given in appendix B. The overall structure is summarised in figure 3.1.

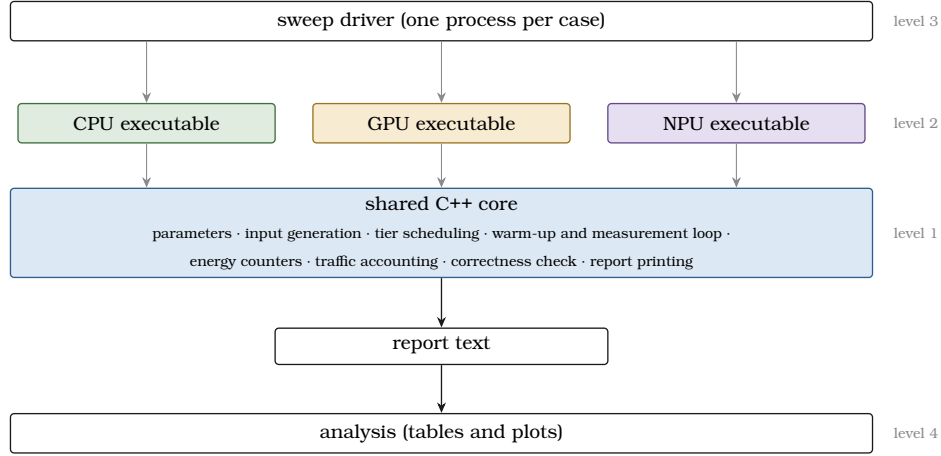


Figure 3.1: The four levels of the measurement harness. The measurement procedure is defined only once, in the shared core of the first level, and the three implementations provide exclusively the function that performs the computation.

3.3 Measured quantities

3.3.1 Measurement scopes: algo and end-to-end

Comparing three architectures requires a decision about what exactly counts as cost. An accelerator that completes the computation very quickly but requires the whole input to be transferred to and from main memory is not necessarily faster than a processor working in place. For this reason each case is measured in two scopes rather than one.

The end-to-end scope covers the whole call from the host’s point of view: everything required so that, given an input in main memory, the result is also found in main memory. The algo scope covers only the region in which the computation itself executes. The difference between the two is the cost imposed by heterogeneity itself, and is summarised per architecture in table 3.2 and schematically in figure 3.2.

Architecture	Content of algo	Difference from end-to-end
CPU	identical to end-to-end	—
GPU	region between CUDA events	H2D/D2H transfers, memory allocations
NPU	region of the offload	device discovery, loading the offload configuration

Table 3.2: The content of the two measurement scopes per architecture.

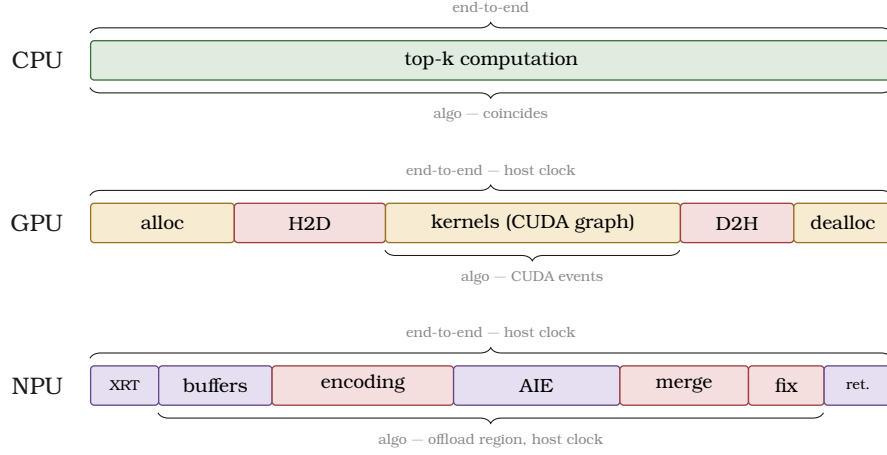


Figure 3.2: The two measurement scopes per architecture. The color of each stage indicates which unit its work belongs to, with red for the extra cost imposed by each architecture: bus transfers for the graphics unit, encoding and merging on the host for the neural processing unit.

For the processor the two scopes coincide by definition and result from the same measurement. This is not a shortcoming of the harness but an accurate description of the architecture: the processor is itself the accelerator, so there is no host cost to subtract. Any other choice would give the processor an advantage it does not deserve.

The two scopes concern more than time. The energy meter opens and closes at exactly the same points in the code as the corresponding timer, so time and energy refer to the same region of execution and are directly comparable to each other. A third, wider window around the entire measurement loop is additionally recorded, which is used as an upper-bound sanity check and is discussed in section 3.3.3.

The choice of scope determines the question a measurement answers. End-to-end captures the total cost of solving the problem and is the scope in which the comparison between the three architectures is fair, because none of them hides its own data transfer cost. Algo isolates the quality of the kernel itself and is useful for comparing implementations within the same architecture. It is not, however, sufficient on its own for claims across architectures, precisely because it omits the transfers.

3.3.2 Timing

The two scopes of section 3.3.1 are not necessarily measured with the same mechanism. The end-to-end scope is always measured with a host clock, on either side of the call, since by definition it concerns time as perceived by the host. The algo scope is measured with whatever mechanism each architecture provides for the region of its own computation.

On the graphics unit the measurement cannot be done with a host clock. Kernels launch asynchronously: the call returns as soon as the kernel is placed in the command stream, not when it completes, so a host clock would measure the time of the launch rather than the time of execution. Forced synchronisation would fix the problem, but would add the cost of the synchronisation itself to the kernel. For

this reason events are used, recorded within the command stream and timed by the device itself.

On the neural processing unit there is no equivalent device-side meter, so the algo scope is defined by a host clock around the region of the offload. This region does not contain only the execution on the tiles: it includes allocating and preparing the buffers, dispatching the jobs, and the final merge of the partial results on the host. This quantity should therefore not be read as the computation time of the AIE tiles, but as the cost of the entire offload procedure. The distinction is essential for the conclusions of chapter 6.

Three quantities per scope are derived from the 50 measured repetitions of each case. The mean expresses the expected cost and is the quantity on which comparisons are based. The standard deviation, computed with the sample count as divisor, expresses their spread under the pinned conditions of section 3.8 and serves as an indication of measurement stability. The minimum represents the best case, that is the repetition least affected by system interference, and is used as a lower bound on the achievable time.

The meters used have a resolution on the order of a nanosecond, while the measured regions range from a few tens of microseconds to hundreds of milliseconds. The cost of the timing itself is therefore negligible compared to the measured interval and is not subtracted.

3.3.3 Energy measurement

Energy is measured by the executable itself rather than by an external wrapper monitoring the process. The reason is that an external tool measures the whole process, whereas the measurement loop occupies only part of it, and that part in fact differs systematically by architecture. Such a tool would not merely add noise, but would introduce a systematic bias in favour of architectures with lower initialisation cost. The meters are therefore read from within the process, at exactly the same points in the code as the timers of section 3.3.2.

The sources of the measurements differ by architecture, but the scope remains symmetric. For all architectures the energy of the processor package is measured via the RAPL counters, and for the graphics unit the power draw of the card is added. This way the end-to-end scope corresponds to the total cost of solving the problem for all three, and the graphics unit does not hide the cost it imposes on the host for transfers and kernel launches.

The reported energy is the sum of the 50 measured repetitions, not their mean. The RAPL counters quantise each individual reading, but accumulate without reset, so the sum remains accurate even when each individual difference is on the order of the quantisation step. Division by the number of repetitions is done afterwards, where a per-operation quantity is needed. Reading the counters requires elevated sudo privileges. When these are not available, the executable explicitly states that the measurement is unavailable rather than reporting zero values as data.

Of the fields exposed by RAPL, only the package field is used. The core subfield was found to be unreliable on this processor: under full eight-thread load, package power

increases by about 53 W, while core records an increase of only 6 W, roughly one-tenth of the actual core consumption. Consequently, the difference between package and core is not a valid way of isolating the neural processing unit’s subsystem from the host cores, and it is not used anywhere in this work.

For the graphics unit, the total-energy counter exposed by the NVML library also proved unsuitable: it advances in steps of about 20 J every few tens of milliseconds, so bracketing a kernel that lasts a few milliseconds returns an arbitrary multiple of the step. Three identical runs gave values that differed by more than a factor of four, while the corresponding time differed by 0.2 %. Instead, a helper thread samples the instantaneous power of the card every 2 ms and integrates it over time.

This approach has a limitation that must be stated. The underlying power reading refreshes far less often than the sampling period, so a single algo window of a few milliseconds relies on a reading that has not had time to refresh. Summed over the 50 repetitions this effect is smoothed out, but the algo/end-to-end split for the card’s energy must be regarded as approximate, unlike the RAPL counters, which refresh on a millisecond scale.

Beyond the gross value, net energy is also reported, that is the gross value minus the energy the system would consume at idle over the same time interval. Idle power is measured before each sweep for the full set of counters of each architecture. The net quantity is preferable especially for the neural processing unit, where the static consumption of the rest of the system dominates, and for the graphics unit, where it cancels out the cost of the sampling mechanism itself.

Finally, it must be stated explicitly what these measurements cannot show. The package meter covers the entire system-on-chip die: the host cores, the neural processing unit’s tiles, the uncore, and the memory controller, all under one shared meter. A run on the neural processing unit is therefore measured as the total energy of the system while it is working, including the preparation of data on the host. This is a fair comparison of the cost of solving the problem and is symmetric across architectures, but it is not sufficient to demonstrate that the unit performs the computation while consuming less energy, because its own consumption is never separated from that of the host. Isolating it would require external metering equipment.

3.3.4 Data traffic measurement

Beyond time and energy, each run also records the number of bytes it moved and the number of operations it performed. These two quantities are the coordinates that place the run on the plot of section 3.4, so an error in counting them does not merely cause imprecision: it shifts the point and can lead to an entirely wrong conclusion about what limits the implementation.

For this reason traffic is not estimated from a model, but accumulated by the code itself at the points where it occurs: per kernel launch on the graphics unit, per group of fused passes on the processor, and per data batch on the neural processing unit.

This distinction is not academic. The obvious estimate, namely the number of elements times the size of the type, assumes that each element is read once. Truncated

bitonic sort, however, traverses the working set repeatedly. The repetition factor is not constant: for k much smaller than n it converges to roughly seven times the naive estimate, while for the larger values of k examined it increases considerably and differs by architecture, as truncation ceases to pay off. In every case it is an order of magnitude away from unity, and using the naive estimate would make every implementation appear to use only a small fraction of the available bandwidth; we would wrongly conclude that none of them come close to the memory ceiling.

The correctness of the count is checked by independent verification where this is feasible. For bitonic sort with k up to 4096, the processor and the graphics unit execute the same stage schedule and compute their traffic in entirely separate code; they nonetheless arrive at the same value down to the last byte. Agreement between two independent counts is strong evidence that the traffic model is correct. For map-reduce the two values differ, and this is expected: although both maintain a local heap per thread, the processor runs only a few threads while the graphics unit runs thousands, with a correspondingly larger working space and an additional tree merge within each thread block. The two implementations genuinely move different volumes of data.

Not every quantity is equally observable, however, and each run states itself the quality of its own numbers. For bitonic sort, both the bytes and the operation count are exact, since the stage schedule is fully enumerable. For map-reduce the bytes are exact, while the comparison count is reported as a lower bound, because the reordering operations of the local heaps are data-dependent. For the ground truth both quantities are lower bounds, since the internal passes of the libraries are not observable from the code that calls them.

Operations are measured in compare-exchanges rather than in floating-point operations. Top- k selection performs no arithmetic operations, so a quantity expressed in FLOPs would not correspond to anything actually executed. The compare-exchange is the natural unit of work for the problem. Mapping these quantities onto performance ceilings is described in section 3.4.

3.4 Roofline model

Execution time says how fast an implementation is, but not why it is not faster. The roofline model [18] answers this question by comparing each measurement against the two upper bounds imposed by the machine: a bandwidth β and a compute capability π . Which of the two binds is a property neither of the machine alone nor of the algorithm alone, but of their relationship, expressed by the operational intensity, the ratio of operations W to bytes Q moved:

$$I = \frac{W}{Q}, \quad P = \min(\pi, \beta I)$$

Figure 3.3 shows how the resulting plot is read.

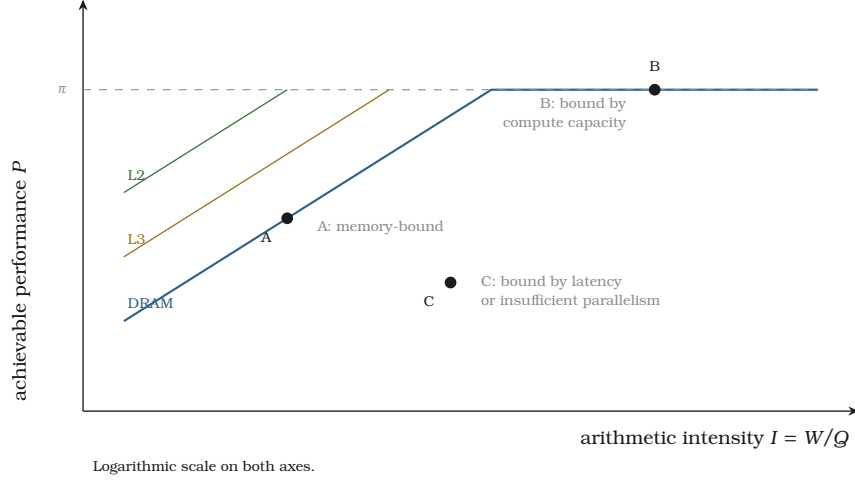


Figure 3.3: Reading the roofline model. The sloped segment corresponds to the bound βI imposed by memory bandwidth, and the horizontal one to the bound π imposed by compute capacity; the achievable performance is the minimum of the two. Because memory bandwidth depends on the size of the working set, the sloped segment is not a single line but a ladder of parallel lines, one per level of the memory hierarchy, and each measurement is compared against the one that corresponds to its own working set. Point A is memory-bound and will benefit only from a reduction in data traffic, B from compute capacity, while C is far from both ceilings: there the limiting factor is not the hardware but the organisation of the execution.

Applying the model in this work departs from the usual practice in three respects, all three necessary for the comparisons to be honest.

First, both ceilings are measured on the same machine with separate executables, rather than taken from manufacturer datasheets. A theoretical figure describes what the hardware could achieve under ideal conditions, whereas what is sought here is what is actually achieved.

Second, the memory ceiling is not a single number. The bandwidth that can be achieved depends on the size of the working set: as long as it fits in some level of cache, data arrives much faster than if it has to come from main memory. For this reason bandwidth is measured over successive working-set sizes, starting at 32 KiB and doubling each time, and each measurement is compared against the ceiling corresponding to its own working set.

This scale measures bandwidth under reuse, and one might object that this is an unsuitable measure for an algorithm that traverses the input only once. In this particular harness, however, the objection does not hold: the input is restored from the backup copy immediately before the timed region, in accordance with section 3.8, so for working sets up to the size of the last cache level the data are already present in it and the measurement corresponds to a second pass. Beyond that size, the copy is not enough to retain them, and the scale converges on its own to the rate of main memory.

Third, the compute ceiling is measured in compare-exchanges, for the reason explained in section 3.3.4. If each compare-exchange uses the result of the previous

one, a dependency chain forms, and what is measured is the latency of a single operation, not the rate at which many can execute; the ceiling would come out several times lower than the true value and every implementation would appear, incorrectly, to be touching it. The measurement kernel therefore runs 64 independent chains in parallel, alternating between three registers so that no operation can be elided by the compiler.

For the neural processing unit the model carries a different meaning, and this must be stated. There is no available AIE kernel with tunable operational intensity, so the ceilings measured concern the unit's feed path rather than the computation on the tiles. They should be read as "how fast can the host prepare the data the unit consumes," which is, in any case, the implementation's actual bottleneck.

3.5 Baseline libraries

The performance baseline is an established library implementation per architecture. A new implementation has value only if it is faster than what a developer already has available.

For the processor, `std::partial_sort` from the standard library is used. Its choice over `std::nth_element` is not self-evident, but for k much smaller than n it is clearly faster: it maintains a heap of size k that stays in cache and rejects most elements with a single comparison.

For the graphics unit, `cub::DeviceTopK` is used, which implements radix selection. It is important that this is a genuine top- k selection function and not a full sort followed by truncation. The comparison is therefore against an opponent that solves the same problem in an optimised way, rather than against an artificially weak baseline.

For the neural processing unit there is no equivalent baseline, and this must be stated explicitly. Whereas a mature library ecosystem with ready-made selection functions exists for the graphics unit, the AIE architecture's toolchain provides a compiler and example kernels, not an algorithms library. That is, no established top- k implementation exists for this hardware itself, either from the manufacturer or from third parties, against which this work's implementation could be compared.

As a baseline, therefore, the same `std::partial_sort` used for the processor is used, which runs entirely on the host and does not make use of the unit at all. This choice changes the question the comparison answers. It does not answer "is our implementation better than an optimised AIE kernel," because no such kernel exists to compare against. It answers "is it worth offloading the computation to the unit, or is it preferable to run it with the means the host already has," which is the question anyone actually considering its use faces in practice.

The fact that the baseline is shared between the two architectures also provides an independent check on the harness. Because it is literally the same code running on the same hardware, the two measurements ought to agree. Their agreement in the results of chapter 7 confirms that the harness does not introduce systematic bias depending on which architecture is declared.

3.6 Correctness checking

A timing measurement is meaningful only if the result is correct. For this reason every case of every sweep is verified, and those that fail are recorded as failures rather than included among the results.

Checking is done against a separate ground truth, which should not be confused with the baseline libraries of section 3.5: those constitute a performance baseline, whereas this one judges whether a result is correct. It is the simplest possible implementation: a copy of the input is fully sorted on the host and the first k elements are kept. It is slow and is not meant to compete with anything else; its role is to be obviously correct, so that its verdict can be trusted.

Comparing the output against this ground truth is not done the same way for all types. For integer types, exact equality is required. For floating-point types, a tolerance is used, relative or absolute depending on the magnitude of the values. The reason is that different architectures perform operations in a different order and with different intermediate precision, so two correct results may differ in the last digits.

When both the full and the truncated network are run, the two results are additionally compared against each other. This is an independent check that truncation does not corrupt the result.

In the main sweep of this work, all 2700 cases were verified with no failures.

3.7 Parameter space

Each run is fully determined by the parameters given on the command line. None are read from a configuration file or an environment variable, so that the description of a measurement is always complete and reproducible; the full list of them is given in appendix B. The sections that follow describe the parameters that are swept and their value ranges.

The input size is defined as a power of two, n equal to 2^q , with q swept from 3 to 24. The upper bound corresponds to about 16.8 million elements, a size large enough that the working set far exceeds the cache and the measurement concerns main memory. The lower bound is not arbitrary: for q smaller than 3 the input has fewer elements than the smallest k requested, so there is no selection problem left to measure.

The parameter k takes the values 8, 256, 4096, and 131072. These values are orders of magnitude apart and cover qualitatively different regimes: from a k so small that the heap fits in a handful of registers, to a k that constitutes a significant fraction of the input itself. Where k exceeds n , it is clamped to n .

3.7.1 Data types

Five data types are swept, given in table 3.3.

Type	C++ mapping	Size	Description
int	std::int32_t	32 bit	signed integer
uint	std::uint32_t	32 bit	unsigned integer
float	float	32 bit	single-precision floating point
double	double	64 bit	double-precision floating point
half	_Float16	16 bit	half-precision floating point

Table 3.3: The data types that are swept.

The type affects the measurement in two ways. First, it determines how many bytes are moved per element: for the same element count, double precision requires four times the data traffic of half precision. The position of the same implementation relative to the memory ceiling therefore shifts depending on the type. Second, it determines the cost of a comparison, which is not the same for all types.

3.7.2 Input distributions

The distribution of the input values is not immaterial. Heap-based algorithms perform work only when a new element exceeds the heap’s root, so the order in which values appear drastically changes their running time. The five distributions of table 3.4 are swept; the executable additionally supports the normal distribution, which was not used in the measurements since trimodal covers the case of clustered values more sharply. The shape of each is shown in figure 3.4.

Distribution	Description
uniform	uniform over the value range
trimodal	three narrow clusters within the range
sorted	ascending order
reverse	descending order
adversarial	values in the lower half with ascending spikes

Table 3.4: The input distributions that are swept.

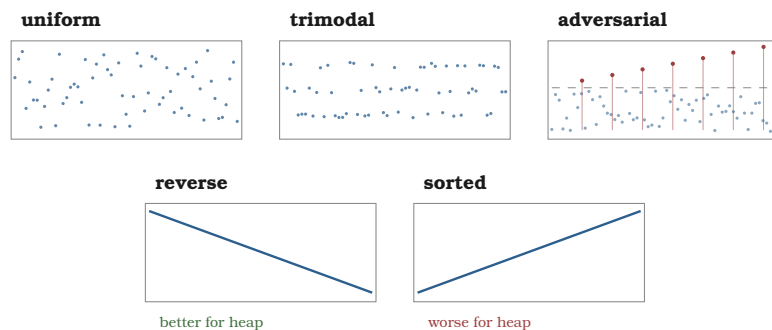


Figure 3.4: The five input distributions, with the value of each element as a function of its position in the array. The uniform and trimodal distributions have no ordering and differ only in which values occur; the two ordered ones are the extremes for heap-based selection. In adversarial the bulk of the values stays below the midpoint of the range (dashed), with isolated spikes at regular intervals.

The two ordered distributions form the two extremes for heap-based selection, for the mechanism of section 2.2.3: descending is the most favourable case and ascending the least favourable. Adversarial was constructed to preserve this unfavourable behaviour without being fully sorted: most of the values lie in the lower half of the range, while spikes of ascending value are interspersed at regular intervals.

Bitonic sort is data-oblivious, since it executes the same comparison network regardless of the values. It therefore serves as a witness: if its time changes across distributions, the change is not due to the algorithm but to the measurement.

3.8 Measurement conditions and reproducibility

The system is not, by default, in a state suitable for measurement: in its default configuration the frequency governor is powersave and frequency boost is active, so the operating frequency varies with load and temperature. This variation is not random noise but a systematic drift during the course of a single sweep, as the system heats up; the standard deviation of samples taken under drifting conditions therefore does not capture the uncertainty of the measurement.

For this reason, before each sweep the system is pinned to a fixed and documented state, summarised in table 3.5. This state is recorded before the sweep and restored after it completes. The corresponding commands are given in appendix B.

Parameter	State during measurement
CPU frequency governor	performance
CPU frequency boost	disabled
CPU threads	8, one per physical core
GPU persistence mode	enabled
GPU power limit	fixed for the whole sweep
Cool-down period between units	15 s
Warm-up repetitions	5 per case
Measured repetitions	50 per case

Table 3.5: The conditions to which the system is pinned before each sweep.

The graphics unit’s power limit is chosen once, within the bounds of table 3.1, and held fixed. The value is arbitrary; what matters is that it does not change either between implementations or during the course of the sweep, since any change to it would simultaneously alter time and energy.

As mentioned in section 3.1, the three units share a common thermal budget. A cool-down period of 15 s is inserted between successive units, so that each does not begin its measurement from the thermal state left by the previous one, while during a sweep the system runs no other computational load. This is a mitigation rather than an elimination of the problem: on a laptop system the thermal coupling between the units cannot be removed, and this constitutes an acknowledged limitation of the measurements.

Each case is run 5 times as warm-up and 50 times under measurement. The mean, standard deviation, and minimum are reported as an indication of the best case.

Before each repetition the input is restored from a backup copy, outside the timed region, so that the cost of the copy is not counted and each repetition starts from an identical input. The second condition is critical: data-dependent implementations would otherwise show artificially improved times on an already partially sorted input.

Beyond the repetitions within a case, each combination of input distribution and random-number-generator seed constitutes an independent sample. The plots in this work aggregate these samples and depict their spread as error bands, so that the uncertainty due to the data can be distinguished from the uncertainty due to the measurement.

Finally, each run itself records the environment in which it executed: the repository revision, the kernel, the compiler, and the driver and runtime library versions of each unit. This information is stored alongside the results, so that each set of measurements documents on its own the conditions of its production and does not depend on external records that might drift out of sync.

Chapter 4

CPU Implementation

4.1 Design decisions

Four decisions run through both implementations in this chapter and are described here once: how vector instructions are exploited, how the same algorithm supports five data types, how half precision is handled, and how work is distributed across threads.

4.1.1 Vectorisation

Both algorithms rely on an operation that vectorises exceptionally well. The compare-exchange of two elements is equivalent to computing their minimum and maximum, and both of these operations exist as instructions that act on an entire vector register at once, without branches. A 512-bit register performs sixteen single-precision integer comparisons in the time of one. The instructions are referred to here by the names of the corresponding intrinsic functions of the standard [19], which is also implemented by the processor described in section 3.1.

The difficulty lies not in the comparison but in positioning the elements. To compare two elements separated by a distance j , they must be brought to the same position in two different registers. When j is smaller than the register width, this is achieved by shuffling within the register: the contents are shifted by j positions, the minimum and maximum are computed, and the result is reassembled by alternately selecting from the two using a mask. The sequence "shuffle, minimum, maximum, select" is the fundamental motif on which the layer fusion of section 4.2 is built, and it is depicted in figure 4.1.

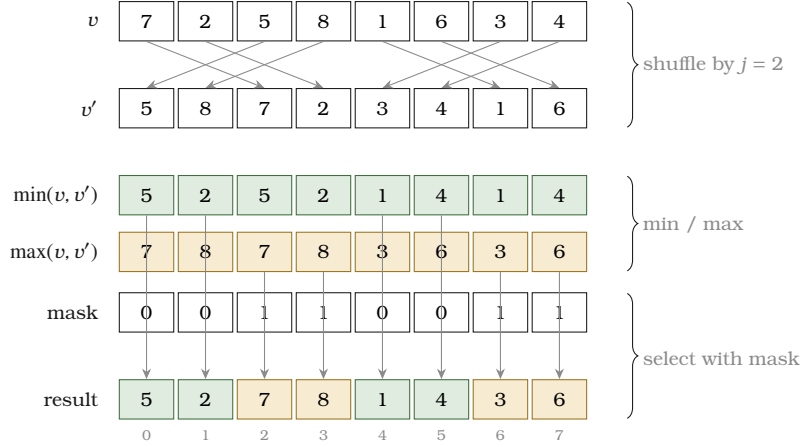


Figure 4.1: The fundamental “shuffle, min, max, select” motif for compare-exchange at distance $j = 2$, in an eight-lane register. The contents are permuted so that each element ends up opposite its counterpart, minimum and maximum are computed simultaneously for all lanes, and the result is reassembled by selecting the minimum where the corresponding mask bit is zero and the maximum where it is one. No branch intervenes and no value is examined individually.

The register width is not known at compile time. The code is compiled with all paths present, and the choice is made at start-up by querying the processor for the instructions it supports. If AVX-512 is available, the 512-bit registers are selected; otherwise the 256-bit ones; and as a last resort the scalar path. The same executable therefore runs on any machine, with the difference that on different hardware it follows a different code path and its results are not directly comparable, as also noted in appendix B.

4.1.2 Vector type abstraction

Vector instructions are specialised by data type: one instruction computes the minimum of signed integers, another of unsigned integers, and another of floating-point numbers. With five types, two register widths, and two algorithms, writing the code directly would lead to twenty nearly identical variants of the same code, with a corresponding maintenance cost and a corresponding risk of one diverging from another.

Instead, a single interface is defined. For each combination of type and width a structure is declared that always exposes the same names: the vector type, its width in elements, load and store, minimum, maximum, masked select, shuffle, and the generation of the candidate mask used by map-reduce. The algorithms are written once against this interface and never refer to a specific instruction.

The benefit is twofold. Adding a data type requires only a new specialisation of the structure and no change to the algorithms. Moreover, the absence of a specialisation is itself detectable at compile time: if no interface has been defined for some combination of type and width, the code is automatically routed to the scalar path instead of failing. The correctness check of section 3.6 validates each combination separately, so the equivalence of the paths is not an assumption but a verified property.

4.1.3 Half-precision floating-point support

Half precision is a special case, because the processor stores 16-bit values but has no arithmetic instructions that operate directly on them at the width the algorithm requires. The obvious solution, scalar processing, would negate any benefit.

The adopted solution exploits the fact that conversion between the two representations is a single instruction. The half-precision specialisation of the interface defines its vector type as that of single precision: the load reads sixteen 16-bit values and converts them to sixteen 32-bit values, all operations execute in single precision, and the store converts them back. The data remain in half precision in memory and are in single precision only for the duration of the computation, as shown in figure 4.2.

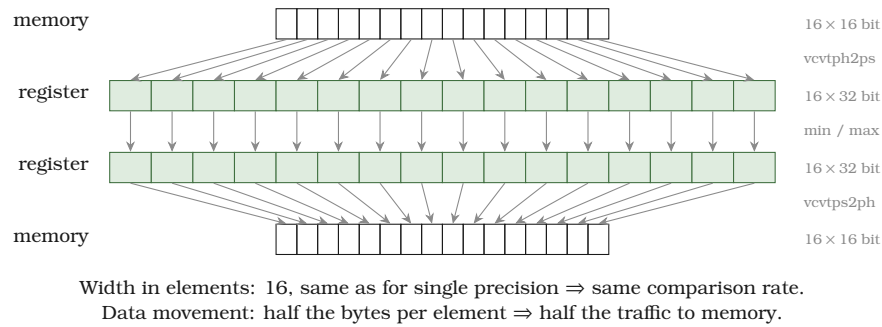


Figure 4.2: The half-precision path. The data remain in 16 bit in memory, where they occupy half the space, and are converted to single precision only within the registers, for the duration of the computation.

This approach is exact rather than approximate. Every half-precision value is represented exactly in single precision, and the only operations performed are minimum and maximum, which select one of the two inputs rather than computing a new value. No rounding intervenes, so the result is identical to that of a hypothetical implementation that operated natively in half precision.

The gain is significant for precisely the reasons discussed in section 4.4. The register width in elements remains sixteen, the same as for single precision, so the rate of comparisons does not change. Data traffic, however, is halved, since each element occupies half the bytes. For memory-bound algorithms, as both were shown to be, this translates directly into a speed-up.

4.1.4 Parallelisation across multiple threads

The two algorithms are parallelised in fundamentally different ways, and this difference determines how aggressively each can be scaled.

Bitonic sort has global dependencies: each group of layers requires the result of the previous one to be complete. Parallelism is therefore confined to within a group, with a synchronisation barrier between groups. The advantage is that threads maintain no private state, and their cost is limited to synchronisation.

Map-reduce has the opposite structure: threads work entirely independently on mutually disjoint segments and meet only once, at the merge. No synchronisation is required during the scan, but each thread maintains its own heap of size k , and the final merge is serial and proportional to the number of threads.

The consequence is that the optimal number of threads is not the same for the two. Bitonic sort tolerates more threads on smaller inputs, whereas map-reduce requires a substantially larger share of work per thread to amortise the cost of the extra heap and the serial phase. In both cases the thread count is dynamically limited according to the input size, so that parallelism never costs more than it returns. The effect of this choice on the results is discussed in section 4.6.

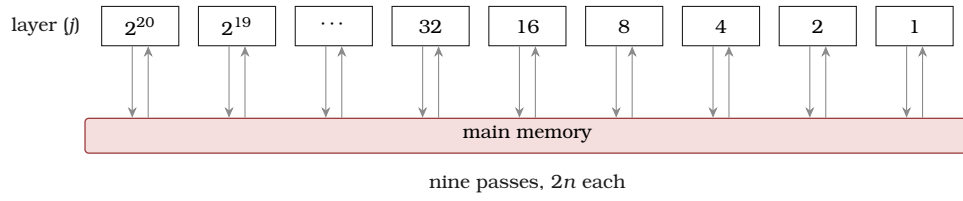
4.2 Bitonic sort

The layer schedule of section 2.2.2 is constructed once and is shared across all three architectures: a first phase forms sorted blocks of size k' , the smallest power of two not less than k , and a second phase merges them in pairs, retaining the best k' out of the $2k'$ compared each time.

A direct translation of this schedule into code results in an implementation that is hopelessly memory-bound. For k equal to 256, the first phase alone consists of 36 layers, each of which would be a separate pass over the entire array. For an input of four million elements this means dozens of consecutive 16 MB reads and writes, with minimal computation in between. The entire implementation of this work is organised around avoiding precisely this pattern.

The basic technique is fusing consecutive layers into a single pass. A layer is characterised by the distance j between the elements being compared, and this distance determines what is feasible. When j is smaller than the vector register width, the two elements of every comparison already lie within the same register. An entire sequence of such layers is then executed as one load, a series of in-register shuffles and comparisons, and one store. When j is larger but the working set of a chunk fits in the cache, the sequence is executed chunk by chunk, with each chunk staying in the cache for all the layers. Only layers with very large j require a separate pass. In addition, a truncation layer whose distance equals the register width also absorbs the re-sort layers that follow it. The two schemes are contrasted in figure 4.3.

(a) naive transfer



(b) layer fusion

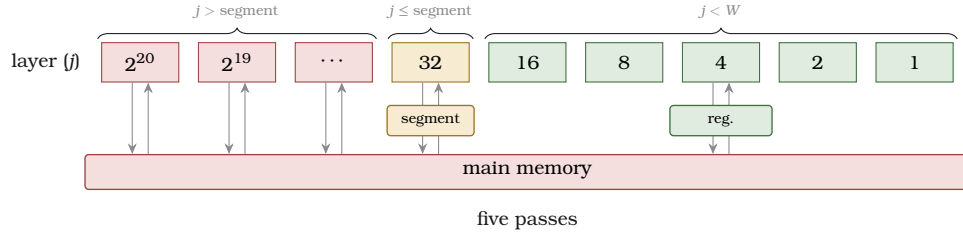


Figure 4.3: The fusion of consecutive layers into a single pass; each pair of arrows is one read and one write of the entire array. In (a) each layer is a separate pass to memory; in (b) the layers are grouped by distance j and only the last group retains a separate pass. The numbers in the figure are illustrative.

The result is that data traffic is not merely measurable, as described in section 3.3.4, but also predictable from the structure of the schedule. The first phase collapses into a single in-place pass, that is, one read and one write per element, hence $2n$. Each step of the second phase reads n_i elements and writes $n_i/2$, and the re-sort that follows adds a further $2 \cdot n_i/2$, for a total of $2.5 n_i$. Because n_i halves at each step, the sum over the second phase tends to $5n$. The total traffic is therefore approximately $7n$ elements, which is the origin of the repetition factor of section 3.3.4. For $n = 2^{22}$ and k equal to 256, this expression gives 117 435 392 bytes, a value that matches the measured one exactly.

Concurrent execution across multiple threads takes place within each group and not between groups, since each group depends on the result of the previous one. The output range of each group is divided among the threads, with boundaries aligned to the chunk size or the cache line so that two threads never write to the same line. A synchronisation barrier intervenes between groups. The thread count is further limited according to the input size, so that for small inputs the cost of synchronisation does not exceed the benefit of parallelism.

Truncation layers cannot be executed in place, because each output element results from two input elements at different positions. For this reason a second array of the same size is maintained, and the two arrays are swapped after each such group. The extra memory is the price paid for avoiding intermediate copies.

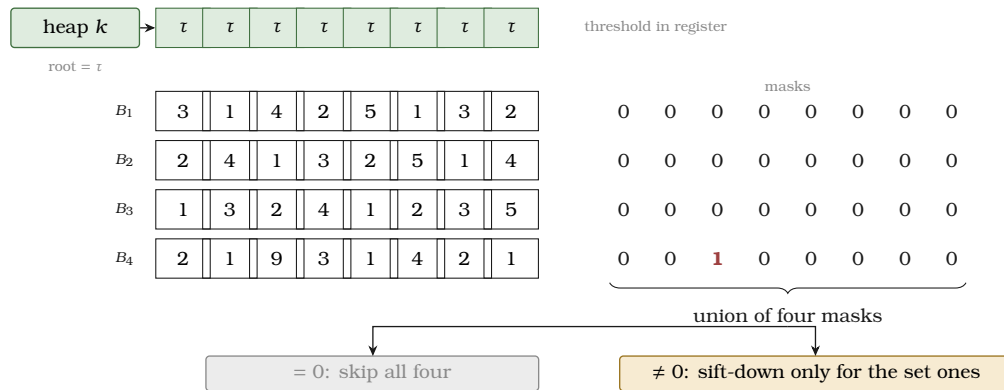
Finally, every vectorised path is accompanied by an equivalent scalar one. If the hardware does not support the required width, or if the parameters of a layer fall outside the range handled by the vectorised implementation, the same sequence of layers is executed in scalar form. The result is identical and only the time differs, which allows the correctness check of section 3.6 to validate both paths.

4.3 Map-reduce with local heaps

The second algorithm follows an entirely different logic from the first. Rather than imposing a fixed sequence of comparisons independent of the data, it maintains the k best elements seen so far and tests every new element against them. The input is divided into segments, one per thread, each thread produces its own k best, and the partial results are merged at the end.

Each thread maintains a heap of size k , whose root is the worst of the k elements selected so far. The heap is bulk-initialised from the first k elements of the segment, which costs linearly in k and is cheaper than k successive insertions. From that point on, the root acts as a threshold: any element that does not exceed it can be rejected without any further work.

The critical point of the implementation is that this rejection happens in bulk. The threshold is broadcast into a vector register and compared simultaneously against an entire block of elements, producing a binary mask that indicates which of them are candidates. The implementation examines four such blocks together and, if the union of the four masks is zero, skips all four with a single check. In the common case, where the heap has already filled with large values and most elements are rejected, the bulk of the input is traversed in exactly this way: it is read, compared vectorwise, and rejected, with no element ever examined individually. The process is depicted in figure 4.4.



In the common case, the left path is followed almost always.

Figure 4.4: Bulk rejection in map-reduce. The root of the heap is broadcast across an entire vector register and compared simultaneously with four blocks of the input, producing one binary mask per block. The union of the masks selects one of the two paths at the bottom of the figure.

Scalar work is performed only for the bits that are actually set. For each such element, the root is replaced and the heap is restored by sifting down. Replacing the root is preferred over extraction followed by reinsertion, since the latter would traverse the heap twice instead of once.

This is precisely where the data dependence described in section 3.7.2 originates. If the input is descending, the heap fills immediately with the largest values, the threshold rises to its maximum possible value from the outset, and the entire re-

mainder of the input is rejected in bulk. If it is ascending, the opposite happens: every new element exceeds the root, no mask is ever zero, and a full sift-down is executed for every element. The same code, given an input of the same size, performs almost no work beyond reading in one case and the maximum possible in the other.

The merge phase is simple and serial. The elements of each local heap are offered in turn to the first one, using the same threshold criterion, and the final result is sorted. The cost of this phase is proportional to the product of the number of threads and k and does not depend on the input size, so for large n it is negligible compared to the scan.

Data traffic follows directly from this structure: every input element is read exactly once, and the merge moves twice the product of the number of threads and k . The figure is therefore exact. The comparison count, in contrast, is reported as a lower bound, precisely because the depth of each sift-down depends on the values; the count records one comparison per element and disregards the reordering of the heap. This is the distinction stated in section 3.3.4.

The thread count is limited far more conservatively than in bitonic sort: roughly one million elements per thread are required. The reason is that every additional thread adds another heap of size k and burdens the serial merge, so beyond some point parallelism costs more than it returns. The consequence is that for an input of four million elements, four threads are used rather than eight, even when eight are requested.

4.4 Memory management and data traffic

The two algorithms solve the same problem on the same machine, but their memory behaviour is radically different. Contrasting them is instructive, because most of the difference in performance is explained by this alone.

The fundamental difference is the volume of traffic. From the expressions in sections 4.2 and 4.3, bitonic sort moves approximately $7n$ elements, whereas map-reduce moves n plus a term proportional to the product of the number of threads and k , which is negligible for small k . For an input of four million integers, the first moves 117 MB and the second 17 MB. The ratio of seven to one is not an artefact of the implementation but of the algorithm itself: the sorting network is obliged to traverse the data many times, whereas heap-based selection traverses it exactly once.

Equally important is the nature of the access. Map-reduce reads the input and never modifies it, so its traffic is purely sequential reading, the most favourable pattern possible for the hardware prefetcher. Bitonic sort, by contrast, writes on every pass, and the comparisons take place between elements separated by a variable distance, which at certain layers exceeds the entire chunk that fits in the cache.

The memory footprint also differs. Bitonic sort requires a second array equal in size to the first, for the reason explained in section 4.2, so it allocates a total of $2n$ elements. Map-reduce allocates only the local heaps, that is, the product of the number of threads and k elements, independent of the input size. For small k this

total is small enough to remain permanently in the fastest levels of the hierarchy, and the only traffic to main memory is the reading of the input itself.

The consequence is that the two algorithms sit at different points relative to the ceilings of section 3.4. Both perform few operations per byte, so neither approaches the compute ceiling, but map-reduce achieves a substantially higher transfer rate than bitonic sort, as expected from its more favourable access pattern. Bitonic sort pays twice over: it moves a sevenfold volume, and it moves it at a lower rate.

It should finally be noted that this advantage is not universal. The term relating to the local heaps grows with k , and for the larger values examined it ceases to be negligible: the heap no longer fits in the fastest cache levels, and the merge phase, being serial, acquires a weight comparable to the scan. Bitonic sort, whose traffic depends much more weakly on k , incurs no equivalent penalty. The comparison of the two across the full range of parameters is carried out in chapter 7.

4.5 Reference implementation

The baseline for the processor is `std::partial_sort` from the standard library, for the reasons explained in section 3.5. It is integrated into the same harness as this work's two implementations, runs in the same measurement loop, and is verified by the same correctness check, so its times are directly comparable.

There is, however, an asymmetry that must be stated explicitly, because it affects the interpretation of every speed-up reported in chapter 7. The library function is strictly single-threaded: it accepts no thread count and makes use of only one core. Bitonic sort, in contrast, uses up to eight threads, and map-reduce up to four or eight, depending on the input size. Every ratio of times between this work's implementations and the library therefore conflates two distinct causes: the difference in algorithm and the difference in core count.

This choice is not arbitrary. The question posed by this work is not which algorithm is better under equal resources, but how much faster the problem can be solved on each architecture relative to what a developer already has available. The established solution for the processor is indeed single-threaded. The reader should nonetheless bear in mind that a thread-for-thread comparison would be stricter and would yield smaller speed-ups.

It should finally be noted that the measured data traffic of the reference implementation is a lower bound, as mentioned in section 3.3.4: only the input and output are counted, while the library's internal passes are not observable. Every memory ceiling utilisation percentage reported for it is therefore an underestimate and is not directly comparable with the corresponding percentages of the other two implementations.

4.6 Limiting factors

Placing the measurements against the ceilings of section 3.4 shows that the limiting factor is not the same across the full range of parameters, and in fact it reverses.

For small k and large input, both implementations are clearly memory-bound. At the largest input examined, bitonic sort approaches two-thirds of the measured bandwidth ceiling and map-reduce a little over half, while both utilise less than five per cent of the compute ceiling. The gap between the two figures leaves no room for doubt about the nature of the limitation.

This diagnosis has a practical consequence for any attempt at improvement. Bitonic sort is not slow because it performs its comparisons inefficiently, but because it moves a sevenfold volume of data. No further optimisation of the compute kernel is going to speed it up substantially; the room for improvement lies exclusively in reducing traffic. Map-reduce, on the other hand, already reads every element exactly once and operates close to the rate at which memory can feed it; under this condition it is essentially optimal, in the sense that it cannot be made faster without reading less.

For large k the picture changes radically, and neither of the two approaches either ceiling. At the largest value of k examined, utilisation of both ceilings falls to figures on the order of one per cent for map-reduce and a little above ten per cent for bitonic sort. When a measurement is far from both ceilings, the limitation is neither bandwidth nor compute capability, but latency and insufficient parallelism.

The mechanism is clear for map-reduce and follows from section 4.3. As k grows, the heap's threshold falls and the rejection rate collapses: the fast path that skips four blocks at once ceases to activate, and almost every element forces a full sift-down. At the same time the heap stops fitting in the fastest cache levels, so every sift-down becomes a sequence of accesses at a random address and a depth logarithmic in k . The algorithm turns from bandwidth-bound to latency-bound, and at the largest value of k it ends up slower than bitonic sort, which it had outperformed by a factor of six for small k .

A secondary but real factor is the thread-scaling limits described in the preceding sections. Because map-reduce requires roughly one million elements per thread, only a fraction of the available cores is used at medium input sizes. Part of the gap from the ceiling in this region is therefore not due to the algorithm but to the fact that the processor is not working at full capacity.

Chapter 5

Implementation on the GPU

5.1 Design choices

Two issues run through both implementations in this chapter: the handling of data types and the mapping of work onto the device's execution structure.

The handling of types follows the same logic as on the CPU, but in a simpler form. A structure is defined per type that exposes the operations the algorithms need, namely comparisons, the minimum, the maximum, and the neutral value used for padding, and the algorithms are written once against it. The essential difference from the CPU case is that here there is no question of selecting a width at run time: the compiler generates code for the specific architecture of the device, so the abstraction serves only to distinguish types and not to distinguish vector extensions.

A mapping between the host's types and the device's types is also required, since half precision is represented by a different type in the two environments. This mapping is performed at a single point and is transparent to the algorithms, which have no knowledge of which side they run on.

5.1.1 Mapping onto the grid and thread blocks

The device executes threads organised into blocks, and blocks into a grid. This distinction is not merely administrative: threads within the same block can share fast memory and synchronise with one another, whereas threads in different blocks have neither capability. Each algorithm must therefore place its dependencies within a block wherever possible.

Both implementations use blocks of 256 threads, a size that balances having enough threads to hide memory latency against not so many that the number of blocks able to coexist on each multiprocessor is restricted.

The way the grid size is determined, however, differs fundamentally between the two, and the difference reflects the structure of each algorithm. In bitonic sort the grid is determined by the data: each block takes on one tile, so the number of blocks is simply the number of tiles into which the active segment is divided. The mapping is direct and leaves no freedom of choice.

In map-reduce, by contrast, the grid is a matter of decision, and the decision follows from three independent constraints. The first is the occupancy the hardware allows, which the implementation queries from the device rather than assuming. The second is the heap-filling condition of Section 5.3.1, which deliberately reduces the number of blocks when more threads would mean heaps that never filter. The third is the available workspace: because each thread requires a heap of size k , the total number of threads is bounded by an upper limit on memory, which becomes binding for large k . Of the three, the strictest prevails.

This last dependency is worth noting, because it runs counter to intuition: the larger k grows, the fewer threads the device can run, precisely because each thread costs more memory. This constraint acts in the same direction as the one in Section 5.3.1, and together the two explain why the implementation is unable to make use of the device at large values of k .

Finally, regardless of grid size, the order in which threads access memory is decisive. Simultaneous accesses by neighbouring threads are coalesced into single transactions only if they touch neighbouring addresses. For this reason, the map-reduce scan does not assign each thread a contiguous segment, as would be natural on the CPU, but strides through the input with a stride equal to the total number of threads.

5.2 Bitonic sort

The schedule of stages is the same as in Section 4.2, since it is built from shared code. What changes is the way it is executed, and the problem the implementation must solve remains the same: the number of stages is large, and the naive mapping of one stage to one kernel launch would mean dozens of successive passes over global memory.

On the GPU a second cost is added that does not exist on the CPU. Every kernel launch carries a fixed overhead, and a scan with hundreds of launches pays that overhead hundreds of times. For this reason the entire network is not launched kernel by kernel; instead it is captured once as an execution graph and then replayed. Scheduling is thus performed once rather than for every stage.

The kernels that make up the graph are of three kinds, and the choice between them is determined by the distance j of the stage, exactly as on the CPU. When the distance is small, an entire sequence of stages is executed inside the shared memory of a thread block, as described in Section 5.2.1. When it is too large to fit within a tile, the data remain in global memory and a single kernel executes up to five successive stages, holding the intermediate results in registers. The truncation stages, finally, have their own kernel, which reads two elements and writes one to a second array, with the two arrays alternating just as on the CPU.

5.2.1 Organisation into tiles and shared memory

The shared memory of a thread block is explicitly managed, in contrast to the CPU's cache, which operates automatically. The implementation loads a tile of data into it,

executes as many stages as possible entirely within it, and writes it back once. Any stages merged in this way incur no additional traffic to global memory.

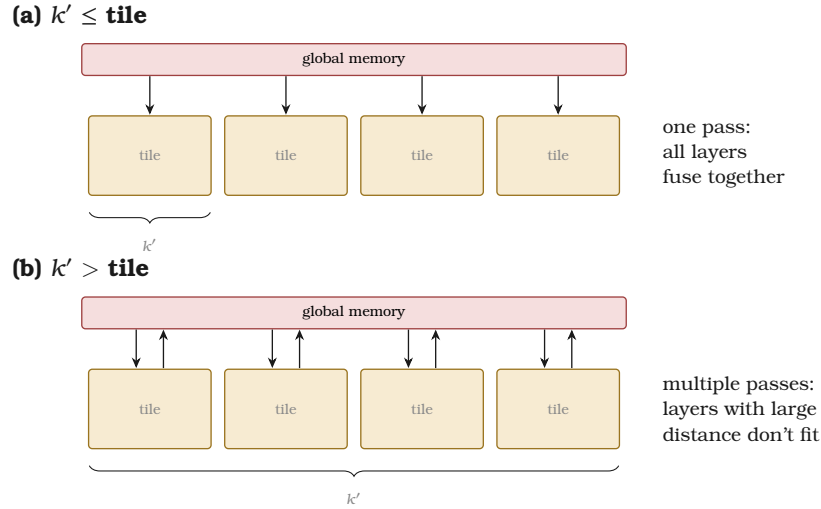
The tile size follows from two constraints. On one hand, it cannot exceed the 48 KB available per thread block without special configuration. On the other hand, it is chosen smaller when the working set fits in the device's second-level cache, in which case the constraint is no longer traffic to main memory but the number of blocks that can be simultaneously active. For 32-bit integers the tile ends up at 8192 elements.

This constraint is not specific to the present implementation: published partial-sorting methods for GPUs are subject to the same bound and do not support k beyond a few hundred, precisely because their structures must fit in shared memory and in registers [3, 5].

This limit has a consequence that determines the behaviour of the whole implementation. A stage can be merged only if the distance between the elements it compares fits within the tile, that is, if it is smaller than half the tile's size. Because the distances of the first phase reach up to half the size of the sorted blocks, the critical size is the k' of Section 4.2: as long as the blocks fit in the tile, almost all stages are merged.

Figure 5.1 contrasts the two cases. For small and medium values of k this holds, and the data traffic turns out essentially identical to that of the CPU: about seven times the naive estimate, with the two architectures agreeing down to the last byte for k up to 4096, as noted in Section 3.3.4. This agreement is not coincidental: the same schedule is executed with an equivalent merging strategy, only in explicitly managed memory instead of automatic.

For the largest value of k examined, this condition breaks down. The sorted blocks reach a size of 131072 elements, sixteen times larger than the tile, so most of the stages of the first phase can no longer be merged and instead execute as passes over global memory. The measured traffic increases correspondingly, from sevenfold to between twenty-five and fifty-six times the naive estimate, depending on the input size. This is the reason why the repetition factor of Section 3.3.4 is stated only for k much smaller than n , and part of the explanation for the implementation's behaviour at large k .



Measured traffic goes from $7n$ to $25n$ to $56n$ once k' exceeds the tile.

Figure 5.1: The relationship between the sorted block and the shared-memory tile. A layer can execute without returning to global memory only if the distance between the elements it compares fits in the tile. As long as sorted blocks of size k' fit in it (a), nearly all layers fuse into one pass and data traffic matches that of the processor. Once k' exceeds the tile size (b), layers with large distance execute as separate passes over global memory and traffic multiplies. For 32-bit integers the tile holds 8192 elements, so the transition falls between the two largest values of k examined.

5.3 Map-reduce with local heaps

The algorithm is the same as in Section 4.3: each thread maintains a heap of size k holding the best elements it has seen, the root acts as a threshold, and the partial results are merged at the end. The difference lies in the scale of the decomposition and its consequences.

On the CPU there are few threads and each one takes on a contiguous segment of the input. On the GPU there are thousands of threads, which stride through the input with a stride equal to their total number, so that simultaneous accesses by neighbouring threads touch neighbouring addresses and are coalesced into single memory transactions.

Each thread maintains its own heap, not one shared per block. The heaps do not fit in shared memory, since the product of the number of threads and k far exceeds the 48 KB available per block, so they are kept in a workspace in global memory. This is exactly the point where the implementation diverges substantially from the CPU one in terms of data traffic: the workspace is proportional to the product of the number of threads and k , and with thousands of threads this term ceases to be negligible.

The merging is performed at two levels. Within each thread block, the heaps are combined pairwise in a tree, with a number of steps logarithmic in the block size.

The result of each block is then merged with the rest. This structure avoids the serial phase that burdens the CPU, but introduces its own cost, since every step of the tree reads and writes heaps from global memory.

5.3.1 Parallelism limit for large k

The algorithm's effectiveness rests entirely on rejection: an element that does not exceed the heap's root costs only a single comparison. The root, however, only becomes meaningful as a threshold once the heap is full. As long as it holds fewer than k elements, every new element must be inserted and no rejection takes place.

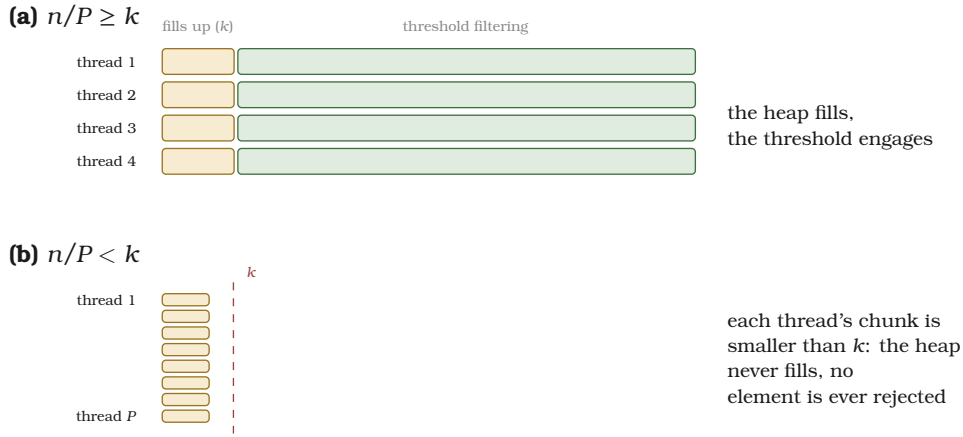
This observation gives rise to a quantitative constraint. If an input of size n is divided among P threads, each thread sees n/P elements and its heap fills only if $n/P \geq k$. This condition is equivalent to

$$P \leq \frac{n}{k}$$

and sets an upper bound on the number of threads that can contribute meaningfully. Beyond this point, additional threads do not speed up the computation but burden it: they fill the workspace with heaps that never filter and add levels to the merge tree.

This constraint is binding precisely where the GPU would be most useful. For the largest value of k examined and an input of sixteen million elements, the ratio n/k is a mere 128. A device with thousands of cores can therefore meaningfully occupy only one hundred and twenty-eight threads, a fraction on the order of one percent of those available. The hardware is underused not because of a weakness in the implementation but because of the structure of the algorithm itself.

Figure 5.2 depicts the two cases. The implementation recognises this condition and reacts to it. Before launch, it is computed whether the heaps manage to fill under full occupancy of the device. When this holds, the number of blocks is deliberately limited, inversely proportional to k , so that each thread takes on a larger share of the input and its heap filters effectively. Nominal parallelism is thus sacrificed for actual rejection. When, conversely, the heaps are not going to fill under any circumstance, reducing the number of blocks would offer nothing, and the device is used at full occupancy.



Useful parallelism is therefore bounded by $P \leq n/k$, regardless of grid size.

Figure 5.2: The parallelism limit of map-reduce for large k . Each thread maintains its own heap of size k and can reject elements only after the heap fills, i.e., only after it has seen at least k elements. When the chunk assigned to each thread is larger than k (a), the threshold engages early and the rest of the chunk is filtered. But when the number of threads increases enough that the chunk becomes smaller than k (b), no heap ever fills and filtering never engages. This is a property of the algorithm, not the implementation: neither the grid size nor the workspace layout can remove it.

5.4 Memory management and data traffic

On the GPU, memory management acquires a dimension that does not exist on the CPU. It is no longer a single hierarchy of successive levels, but two separate memory systems connected by a bus incomparably slower than either. The measured local memory of the device delivers about 320 GB/s, while the bus to the host delivers a mere 14 GB/s: a difference of more than twentyfold.

Within the device, the traffic picture is analogous to that of the CPU but not identical. For small k , bitonic sort moves a sevenfold volume, as on the CPU, while map-reduce moves roughly threefold. The difference is due entirely to the heap workspace described in Section 5.3: with thousands of threads, each maintaining its own heap in global memory, the term that was negligible on the CPU becomes comparable to the scan itself.

It is the transfer to and from the device, however, that dominates. The input must cross the bus once toward the device, and the result must return. For an input of four million integers, the roughly seventeen MB require more than one millisecond just to arrive there, while the bitonic sort kernel itself completes in half that time. The transfer, in other words, costs more than twice the computation it serves.

This observation is precisely why the work measures two scopes rather than one, as defined in Section 3.3.1. For the same run, the algorithmic time is about half a millisecond while the total exceeds two: more than three-quarters of the cost has nothing to do with the computation at all. A comparison based solely on the algorithmic scope would credit the GPU with an advantage it does not have once the

problem is framed as data transfer to and from main memory.

It is worth noting that the established technique for improving transfers, namely the use of pinned host memory to avoid an intermediate copy, offers no substantial benefit on this particular system. Measurements of the two paths differ by a margin on the order of two percent, which indicates that the limiting factor is the bus itself and not the way data is prepared on either side of it.

As for the memory footprint on the device, the two implementations follow the same logic as their CPU counterparts. Bitonic sort allocates two arrays of size n , since the truncation stages are not performed in place. Map-reduce allocates the input plus the heap workspace, which here, however, is not negligible: it is proportional to the product of the number of threads and k , and for large values of k it can approach the size of the input itself.

5.5 Ground truth

The baseline for comparison on the GPU is the top- k selection function of the CUB library, which ships with the CUDA toolkit. Unlike the CPU case, where the ground truth is single-threaded and the comparison conflates two causes, here there is no such asymmetry: the library runs on the same device, exploits the same parallelism, and is measured in the same way. The comparison is therefore rigorous.

Its integration follows the two-call protocol the library requires. The first call performs no computation but reports how much auxiliary space is needed, which is then allocated, and the second call performs the selection. Only the second is bracketed by the events that define the algorithmic scope, while allocation and transfers are left outside it, exactly as in the two implementations developed in this work.

The recorded data traffic is limited to the input and output and is reported as a lower bound, since the library's internal passes are not observable from the calling code. The same holds for the number of comparisons.

It should be stated from the outset that this baseline is a strong one. On the largest input examined, the library completes the selection in a time that remains nearly constant as k varies over four orders of magnitude, while both implementations developed in this work are burdened significantly. Only for the smallest k does map-reduce edge it out, marginally. The interpretation of this difference is given in Section 5.6, once the mechanism behind it has first been explained.

5.5.1 Radix selection in CUB

The library [14] uses neither a sorting network nor heaps, but the radix selection of Section 2.2.4.

Two properties of the method explain its performance. First, it never sorts the entire input; it performs a few passes, each of which is a sequential read accompanied by a frequency count, that is, exactly the pattern the architecture executes best. Second, and more importantly, the number of passes is determined by the width of the data type and not by k . Selecting eight elements and selecting one hundred and thirty thousand cost essentially the same.

This independence from k is the fundamental difference from both algorithms developed in this work. Bitonic sort is burdened as k grows, because the sorted blocks grow and, as shown in Section 5.2.1, cease to fit in the tile. Map-reduce is burdened even more, for the reason given in Section 5.3.1. Radix selection has no equivalent breaking point.

The method is not, however, indifferent to the data. The rate at which the search narrows depends on how uniformly the digit values are distributed: when many elements share the same significant digits, the critical bucket remains large and more passes are required. This dependence is of a different nature from that of the heaps, since it concerns the representation of the values rather than the order in which they appear, and it resurfaces in the discussion of data types in Chapter 7.

5.6 Bottlenecks

On the GPU, bottlenecks must be examined separately for each measurement scope, because they are not the same. In the end-to-end scope, the bus to the host dominates, as shown in Section 5.4: more than three-quarters of the time has nothing to do with computation at all. What follows therefore concerns the algorithmic scope, where the limits of the device itself become visible.

For small k , bitonic sort is clearly limited by memory, reaching about two-thirds of the measured memory bandwidth ceiling while exploiting only about two and a half percent of the computational one. The implementation, in other words, exploits the device efficiently. Its problem is not the way it moves data, but the fact that it is forced to move a sevenfold volume.

The contrast with the ground truth library makes this distinction quantitative, and it is the most instructive point of the chapter. The library exploits only about one-quarter of the bandwidth ceiling, clearly less than bitonic sort, and yet finishes faster. The reason is that it moves a volume equal to the naive estimate, rather than a sevenfold one. The ratio of the two times predicted from these two pairs of quantities coincides almost exactly with the one measured.

This conclusion is worth stating explicitly, since it is easy to overlook: closeness to the ceiling measures the quality of the implementation, while the total traffic volume measures the quality of the algorithm. An implementation can exploit the hardware better than another and still lose, if its algorithm demands more from the hardware. This is exactly what happens here.

Map-reduce for small k is the only point where an implementation developed in this work outperforms the library, although marginally. It reads the input once and rejects almost everything, so it approaches the minimum possible work for the problem. Utilisation of the ceiling remains modest, which indicates that the remaining margin there is not algorithmic but concerns the organisation of accesses.

For large k the picture changes radically. Map-reduce collapses to the point where utilisation of both ceilings becomes a fraction of one percent. When a measurement falls this far from every ceiling, the constraint is neither memory nor computation; it is the absence of work to perform. The reason was given in Section 5.3.1: on the

largest input and for the largest k , only one hundred and twenty-eight threads can filter. The entire device waits on a fraction of one percent of its cores.

Bitonic sort is burdened as well, but far more mildly. Utilisation of the memory ceiling drops to roughly one-third, as the loss of merging in shared memory increases traffic without a corresponding increase in useful work. It nevertheless remains functional and, for the largest k , ends up two orders of magnitude faster than map-reduce.

Overall, neither implementation is limited by a factor that could be lifted through further code optimisation. Bitonic sort would benefit only from a reduction in traffic, that is, from a different network. Map-reduce would benefit only from a different decomposition for large k , so that parallelism would not depend on the ratio n/k . In both cases the limit is algorithmic rather than implementational.

Chapter 6

Implementation on the NPU

6.1 Design choices

The neural processing unit differs from the two previous architectures not in degree but in kind. It is not a device on which a program runs in the usual sense. It is an array of tiles configured in advance, with no cache, no dynamic memory allocation, and no ability to change its dataflow at runtime.

The design question therefore changes fundamentally. On the processor and the GPU, the question was how the algorithm is distributed across threads. Here, the question is what small, fixed unit of work the array can execute efficiently, and how the host can feed it without interruption. As will become clear, the second part proves harder than the first.

6.1.1 Development flow with MLIR-AIE

No programming environment equivalent to that of the GPU exists. This work relies on the MLIR-AIE framework [13] and the accompanying IRON interface [20] that comes with it. Development requires two separate artefacts, produced by different toolchains and combined at the end.

The first is the description of the array itself: which tiles are used, which data channels connect them, and the pattern by which data moves to and from main memory. This description is written in a domain-specific language embedded in Python, translated into an intermediate representation, and finally into a binary hardware configuration file. The second is the compute kernel executed by each tile, written in C++ using the architecture’s vector primitives and compiled separately.

The consequence that governs everything else is that this configuration is fixed at compile time. The topology, the sizes of the intermediate buffers, and the data-transfer pattern are all decided before each run and do not depend on the problem’s parameters. Changing k or the input size changes nothing on the device; it only changes how the host splits the input and interprets the results.

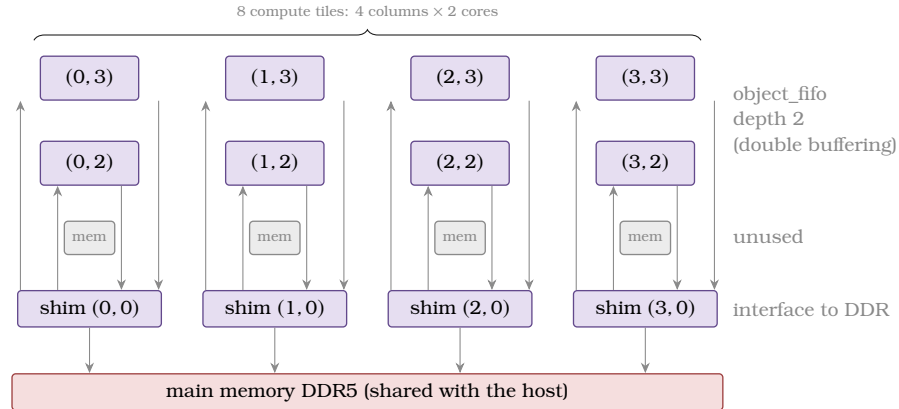
6.1.2 Tile topology and object_fifo

The array is organised into columns. Each column has one interface tile that communicates with main memory and, above it, compute tiles. The bitonic sort implementation uses four columns with two compute tiles each, eight in total.

Data transfer is not programmed with explicit copy instructions but declared as a channel between two tiles. Each channel is backed by a number of intermediate buffers, set here to two. This choice is what enables overlap: while a tile processes one buffer, the direct memory access mechanism fills the other. Without it, each tile would stall on every transfer.

Each compute tile has its own pair of channels to its column's interface tile, one for input and one for output. There is no communication between compute tiles, which is not a limitation of the implementation but a deliberate choice: as will be shown in section 6.2, the work assigned to the array is independent per tile by construction and requires no exchange.

The overall organisation is shown in figure 6.1. The amount of data the array processes in a single dispatch follows from the product of the above. Each tile receives chunks of 1024 elements and processes 128 such chunks per dispatch, so with eight tiles the batch reaches roughly one million elements. Inputs larger than this are split into successive batches.



Per core: 128 chunks $\times$ 1024 int32 elements. Per batch: $8 \times 128 \times 1024 = 1\,048\,576$ elements.

Figure 6.1: The tile topology of the implementation and the object_fifo queues that connect them. Each column has one interface tile (shim) that communicates with main memory and two compute tiles, each of which connects to its column's shim through a pair of depth-two queues: one for input and one for output. The depth of two allows the core to process one chunk while the next is transferred. The memory tiles in the middle row are unused, since each core consumes a portion of the input disjoint from the others and no sharing or redistribution of data between them is required.

6.2 Bitonic sort

An essential difference from the two previous chapters must be stated up front here, as it affects the interpretation of every measurement: the array does not execute the

level schedule of section 4.2. That is, it does not perform truncated bitonic sort.

The reason is structural. The schedule requires comparisons between elements that are arbitrarily far apart, hence data exchange between tiles and multiple passes over the entire input. The topology of section 6.1.2 offers neither: the tiles do not communicate with one another, and every element passes through the array exactly once.

The implementation therefore reframes the problem. The array takes on a fixed, local task: it independently sorts successive sequences of sixteen elements. The host then performs the selection, merging these sorted sequences into a heap of size k .

This division of labour is not arbitrary. Because each sequence comes back sorted, the host can abandon it as soon as it encounters an element that does not exceed the heap's root: everything that follows in the same sequence is necessarily worse. The sorting performed by the array is therefore not an end in itself, but a mechanism that lets the host discard sixteen elements at a time instead of one.

6.2.1 Compute tile kernel

Bitonic sort on an AI Engine tile is also the subject of an official platform example [17], which, however, performs a complete sort on a single tile.

The size sixteen is not arbitrary: it is exactly the number of 32-bit integers that fit in a vector register of the architecture. Sorting sixteen elements with a bitonic network requires ten compare-exchange steps, each of which executes entirely within the register.

Each step follows the same pattern as in section 4.1.1: a shuffle brings each element next to its partner, the minimum and maximum are computed, and a mask selects which of the two each position keeps so as to produce the correct direction. The entire network executes without any access to the tile's memory: the data is loaded once, undergoes ten register-level transformations, and is stored once.

Each kernel invocation processes a chunk of 1024 elements, that is, sixty-four independent sequences. These sequences have no dependencies between them, which allows the compiler to pipeline them and hide the latency of the individual instructions.

6.2.2 Scaling to multiple tiles

Because the sequences are independent, scaling to more tiles is trivial: it suffices to give each tile a different chunk of the batch. Neither synchronisation nor data exchange is required, and the only thing that changes is the transfer pattern declared in the device configuration.

This capability was put to use. The implementation was extended to use two compute tiles per column instead of one, reaching eight in total. The extension produced a noticeable improvement, confirming that up to that point the implementation had indeed been compute-bound on the array.

The more important result of the extension, however, was not this improvement in itself, but that it shifted the bottleneck elsewhere. With eight tiles the array stopped being the bottleneck, and, as quantified in section 6.4.1, time is now spent almost entirely on the host. Further scaling to even more tiles would have had no effect.

6.3 Map-reduce with local heaps

The second algorithm follows the same work-distribution principle, with an even more limited role for the array. Four compute tiles are used, one per column, and the kernel does not sort: for each chunk of the input it computes a summary, namely the best value it contains.

The host maintains the heap of size k and uses the summaries as a filter. If the best value of an entire chunk does not exceed the heap's root, the chunk is rejected in its entirety without examining any of its elements. Only for chunks that pass the filter does the host traverse the individual elements.

The division of roles is therefore the reverse of the intuitive one: the array does not perform the selection but produces the information that lets the host perform it cheaply. This pattern, in which the accelerated part is a filter rather than the computation itself, matches the nature of the array: a fixed, local, fully predictable dataflow operation.

6.4 Memory management and data traffic

The neural processing unit has no memory of its own, as noted in section 3.1. Data does not cross a bus to a separate device, as with the GPU, but already resides in the same physical memory as the host's. What is called a transfer is in fact the synchronisation of a shared buffer.

This difference at first appears to be an advantage, and in terms of transfer cost it is: there is no twenty-fold bandwidth gap as in section 5.4. But it also has the reverse side. Because the memory is shared, every byte the host prepares for the array consumes the same memory bandwidth the processor would use if it performed the computation itself. Preparation is not free, and, as will become clear shortly, it is the dominant cost.

6.4.1 Pipelining and the role of the host

So that the array does not sit idle between successive batches, execution is organised as a three-stage pipeline with two alternating sets of buffers. While the array processes one batch, the host prepares the next and merges the results of the previous one.

The structure of the pipeline is given in figure 6.2. Measuring the individual stages is the most illuminating result of the entire chapter. For an input of sixteen million integers, split into sixteen batches, time is distributed as shown in table 6.1.

Stage	Time (ms)	Percentage
Batch preparation	4.86	61 %
Result merging	2.19	27 %
Waiting for the array	0.54	7 %
Dispatching work	0.11	1 %
Initialisation	0.08	1 %
Total	7.98	100 %

Table 6.1: Time breakdown of the offload region by stage.

The conclusion is unambiguous. Less than one tenth of the time is spent waiting for the array; roughly ninety percent is work performed by the host itself. A pipeline performs only as well as its slowest stage, and the slowest stage is not on the device.

This also explains why, for this architecture, the algorithmic-scope measurement must not be read as the tiles’ compute time, as already noted in section 3.3.2. The scope encloses the entire offload procedure, most of which executes on the processor.

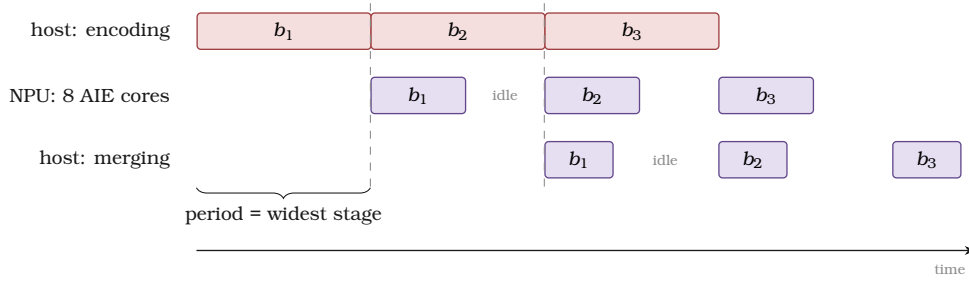


Figure 6.2: The three-stage pipelining of the offload. While the tiles process batch b_i , the host is already encoding b_{i+1} into the second buffer and merging the results of b_{i-1} . Because the three stages overlap, the total time is not their sum but the product of the number of batches and the duration of the widest stage. In this implementation that stage is host encoding: only it executes without interruption, while the other two remain idle for part of each period. No further speedup of the AIE core can therefore reduce the total time.

6.4.2 Data encoding and types

The array’s kernel operates exclusively on 32-bit integers. It does not know, and does not need to know, what the values it compares represent. Every data type used in this work must therefore be mapped to an integer key that preserves the relative order of the values, following the platform’s type-conversion rules [21].

The mapping differs by type. Signed integers are used as is. For unsigned integers, the most significant bit is flipped. For single-precision floating-point numbers, either the sign bit or the remaining bits are flipped, depending on the sign, so that the binary comparison coincides with the arithmetic one. Half precision is first widened to single precision, a conversion that is exact, and then follows the same path.

The critical point is that this encoding executes on the host and forms part of the preparation stage in table 6.1. The cost of the data type on this architecture is therefore not a cost of the device but of the processor. This observation has a practical consequence: the half-precision encoding, initially performed in scalar fashion, proved expensive enough to noticeably burden the dominant stage, and was replaced with a vectorised path that converts eight values per instruction.

6.4.3 Final correction on the host

Double precision does not fit into a 32-bit key. The encoding keeps the 32 most significant bits of the full 64-bit key, so two different values can map to the same key. The array therefore does not return the exact result, but something weaker: a threshold.

Correctness is restored with an additional stage on the host. The worst of the keys returned by the array yields the threshold. The host then traverses the entire input again, collects the elements whose key falls within the threshold, and performs exact selection among these alone.

The cost is one additional pass over the input, which burdens only double precision. The benefit is that the result is exact and passes the same correctness check as all other types, with no relaxation of the criterion. The alternative, that is, accepting an approximate result for a single type, would make the comparison across types unfair.

6.5 Ground truth implementation

The point of comparison is the same standard-library function used for the processor, executed entirely on the host. The reasons for this choice, and what it means for the interpretation of the results, were discussed in section 3.5 and are not repeated here.

It is only worth reiterating the essential point: no top-k implementation exists for this array against which the present work could be compared. The question the comparison answers is therefore not whether the implementation is better than an optimised AIE kernel, but whether offloading the computation to the unit is worthwhile at all, rather than executing it with the means the host already has.

6.6 Limiting factors

The limiting factor has already been quantified in table 6.1 and leaves no room for interpretation: the implementation is bound by the host, not by the device. The remaining question is why the host's work is so expensive.

The preparation stage, which consumes three fifths of the time, involves one full read pass over the input, converting each element to an integer key, and a write of equal volume to the shared buffer. This amounts to data traffic of at least twice the input size, over the same memory and with the same bandwidth the processor would use. The merging stage adds yet another read pass over the results.

This finding leads to the central conclusion of the chapter. Even if the array completed its own work instantaneously, the total time would decrease by less than one tenth. Speedup cannot come from the device, because the device is not where the time is spent. No addition of compute tiles, no optimisation of the kernel, and no increase in frequency would materially change the outcome.

The comparison with the processor sharpens the argument. The host feeding the unit is the same processor that, in chapter 4, solves the entire problem on its own. Here it is asked to perform nearly the same amount of data traffic merely to prepare the data for someone else. It is therefore unsurprising that offloading does not pay off: the cost of preparation is comparable to the cost of the solution itself.

Finally, it must be stressed that the conclusion concerns this particular problem and not the architecture in general. Top-k selection has minimal computational intensity: it performs one comparison per element and reuses almost nothing. A dataflow array performs well when every element that enters it undergoes many operations, so that the cost of feeding it is amortised. Here the opposite holds, and this mismatch, not any weakness of the implementation, is the underlying cause.

Chapter 7

Results and evaluation

This chapter presents the measurement results and interprets them in terms of the mechanisms described in the three preceding chapters. Unless stated otherwise, the plots concern 32-bit integers and a uniform input distribution. The effect of data type and distribution is examined separately in sections 7.6 and 7.7.

7.1 Correctness

A precondition for any timing comparison is the correctness of the result. The main sweep comprises 2700 cases, covering all three architectures, the three algorithms, the five data types, input sizes from 2^3 to 2^{24} , and the four values of k . All were verified against the ground truth of section 3.6, and all passed.

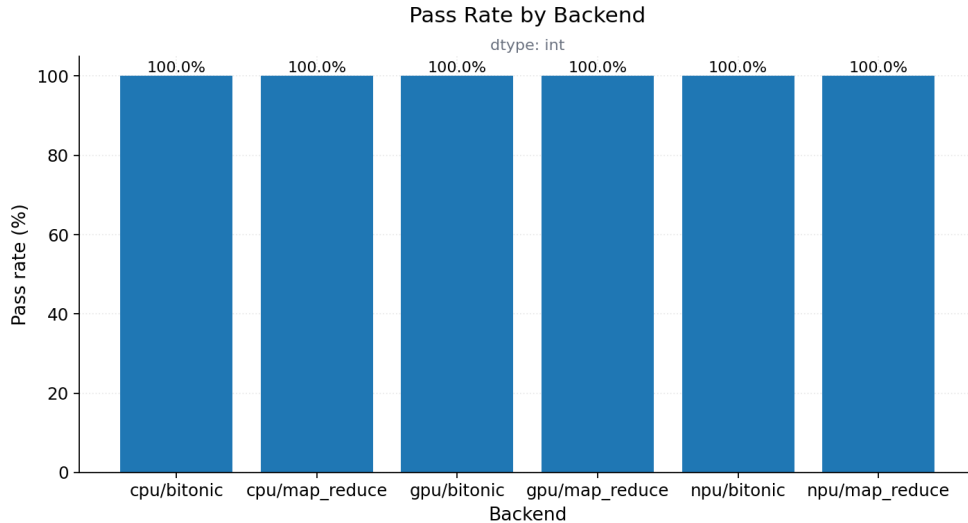
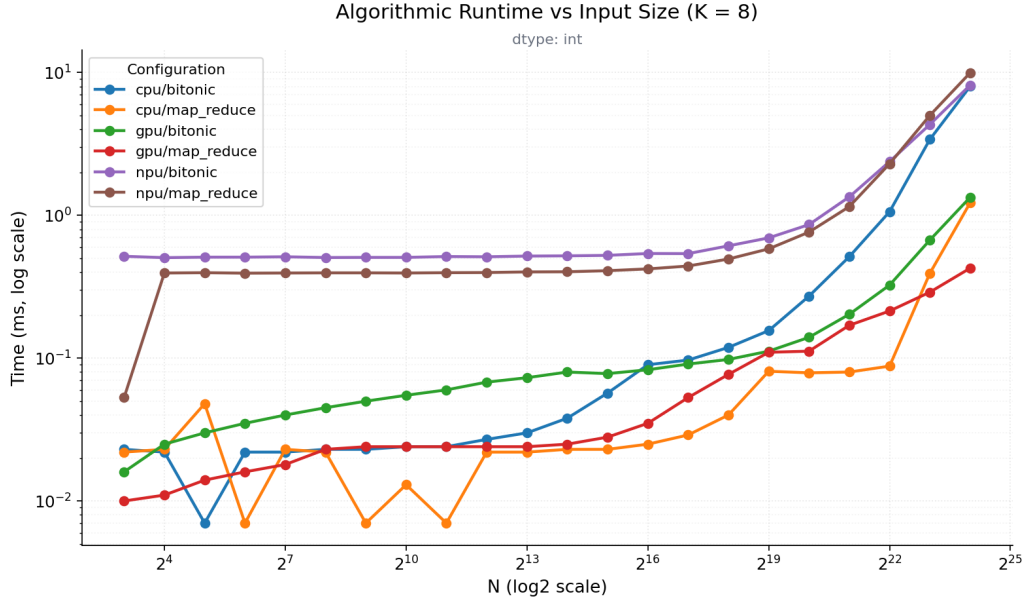


Figure 7.1: Pass rate of the correctness check for the two algorithms of this work, by architecture.

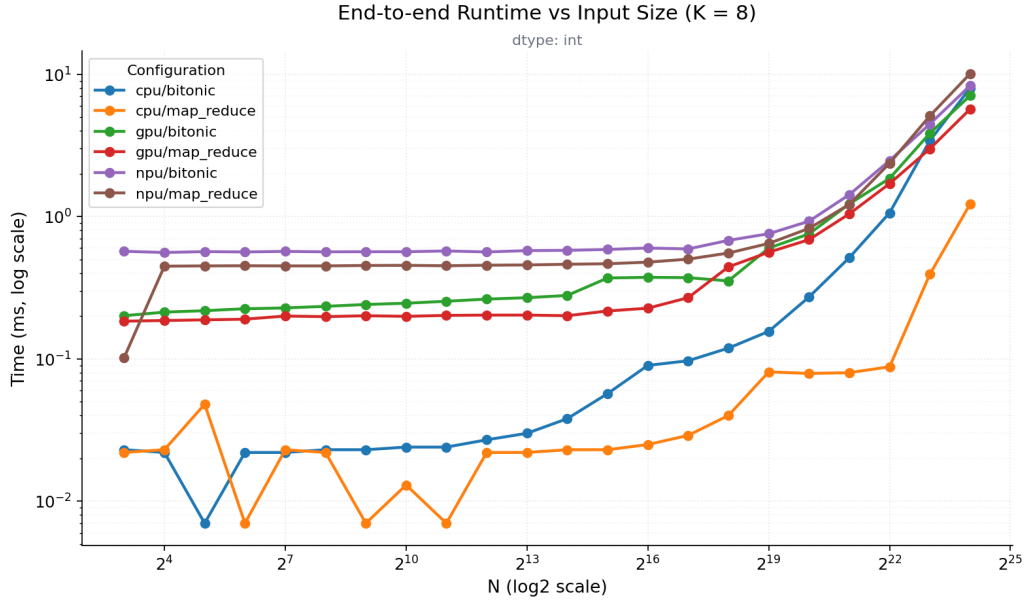
This result is not a procedural detail. It means that every time reported in what follows corresponds to an implementation that produced the correct set of k elements, and that the comparisons that follow indeed concern the same computation.

7.2 Execution time

7.2.1 Dependence on input size



(a) Algorithmic scope.



(b) End-to-end scope.

Figure 7.2: Time as a function of input size, for k equal to 8, under the two measurement scopes. The ranking of the architectures reverses completely between the two.

In the upper figure, which concerns the algorithmic scope, the ranking is not one-sided. The GPU is fastest at the smallest input, completing in 0.010 ms against 0.022 ms for the CPU, and again at the largest, in 0.43 ms against 1.23 ms; for nearly the entire range in between, the CPU is the faster of the two, since its persistent

thread pool now removes per-call overhead that used to dominate at these sizes. The exact size at which the two swap places is itself unstable across repeated runs of the same sweep: at these small absolute times, both architectures sit close enough to the timer’s resolution that the ranking there reflects measurement noise rather than a fixed architectural difference.

The other two architectures show a clear floor at small sizes. The NPU curves remain completely flat around half a millisecond up to about 2^{16} elements: throughout this range the time does not depend on the input size at all, since it is dominated by the fixed cost of dispatch and waiting. The CPU shows an analogous but much lower floor, on the order of five hundredths of a millisecond.

This picture, however, is completely reversed in the lower figure, once the measurement includes data transfer as well. In the end-to-end scope the CPU is faster than every other architecture at *all* sizes examined, including the largest: the GPU kernel’s 0.43 ms becomes 5.69 ms in total, while the CPU remains at 1.23 ms, since for it the two scopes coincide.

This contrast is the first appearance of a pattern that recurs throughout the chapter. The algorithmic scope measures how well each device performs the computation; the end-to-end scope measures whether it is worth delegating to it in the first place. The two answers do not coincide, and section 7.5 shows that it is the second that also determines the energy picture.

The NPU requires 9.97 ms at the largest input, i.e. more than any other architecture under either scope, for the reasons given in section 6.6.

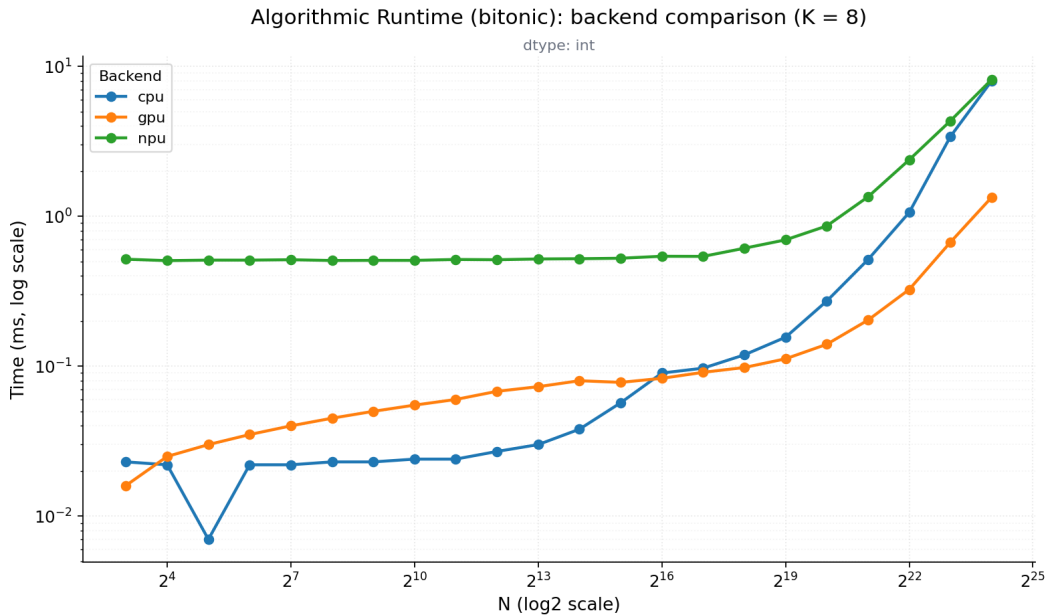


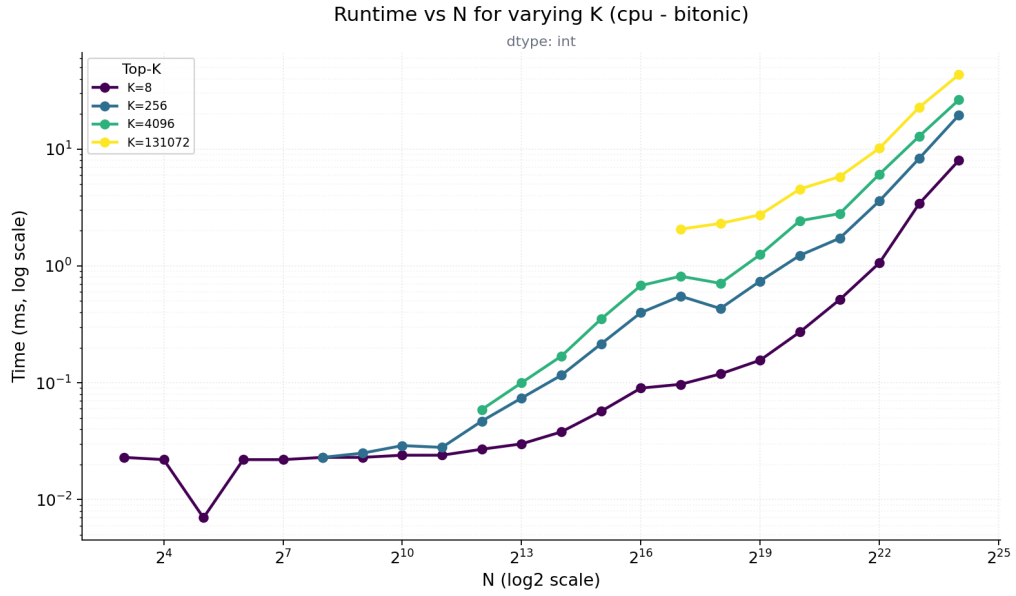
Figure 7.3: The same algorithm across the three architectures, for k equal to 8.

Comparing the same bitonic sort across the three architectures is illuminating, because the same level schedule is executed and the difference is due solely to the implementation. The two general-purpose architectures trade places along the range: the CPU is faster than the GPU through most of the small-to-medium sizes, since it

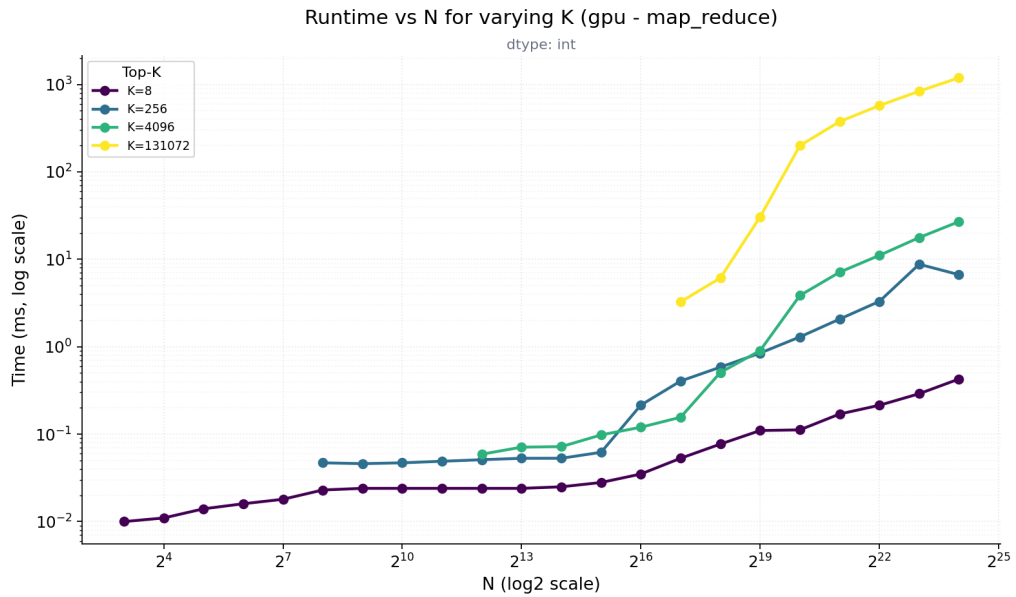
now amortises its fixed per-call overhead well there, and the GPU pulls ahead only at the largest sizes, reaching a factor of six at the largest size examined, consistent with the difference in available bandwidth. As with the comparison above, the exact size at which the two swap places moves between repeated runs of the same sweep and should not be read as a fixed threshold. The NPU sits close to the CPU at these larger sizes, which is consistent with the finding that its time is spent almost entirely on the host.

7.2.2 Dependence on the parameter k

The dependence on k is the point at which the ranking of the algorithms reverses, and is perhaps the most significant finding of this work.



(a) Bitonic sort, CPU.



(b) Map-reduce, GPU.

Figure 7.4: Time as a function of input size, with a separate curve for each value of k . The bitonic sort curves remain close to one another, while the map-reduce curves diverge by orders of magnitude.

The juxtaposition in figure 7.4 captures the phenomenon directly. For bitonic sort the curves of the four values of k nearly coincide, since k affects only the size of the sorted blocks. For map-reduce on the GPU the same curves are separated by three orders of magnitude.

Implementation	k=8		k=131072	
	ms	rank	ms	rank
GPU map-reduce	0.43	1	1191.37	6
GPU bitonic	1.34	3	8.22	1
CPU map-reduce	1.23	2	60.37	5
NPU bitonic	8.17	5	56.70	3
CPU bitonic	8.01	4	43.33	2
NPU map-reduce	9.97	6	59.69	4

Table 7.1: Algorithmic time for the two extremes of k , at an input of 2^{24} elements. The ranking concerns the six implementations of this work and does not include the ground-truth implementations.

The fastest implementation for small k becomes the slowest for large k , a difference of three orders of magnitude. GPU map-reduce slows down by a factor of 2800 between the two extremes, whereas bitonic sort on the same architecture slows down by barely a factor of six. At small k the two CPU implementations now also rank ahead of their position at large k , map-reduce overtaking GPU bitonic and bitonic overtaking NPU bitonic, but the reversal itself is unaffected: the implementation cheapest at k equal to 8 is never the implementation cheapest at k equal to 131072.

The reversal is not a peculiarity of this implementation. Independent studies on GPUs from two different vendors report the same dependence on k and address it with adaptive algorithm switching [3, 15]; there, the growth in time of partial-sorting methods with k is attributed explicitly to the $O(\log^2 k)$ complexity of the underlying bitonic network.

The cause was given in section 5.3.1: the heap only filters once it is full, so the number of threads that contribute is limited to the ratio n/k . For the sizes in table 7.1 this ratio is 128, i.e. the device effectively employs one hundred and twenty-eight threads out of the thousands it has available. The CPU, with only eight threads, never runs into this same limit, and for that reason its slowdown is markedly milder.

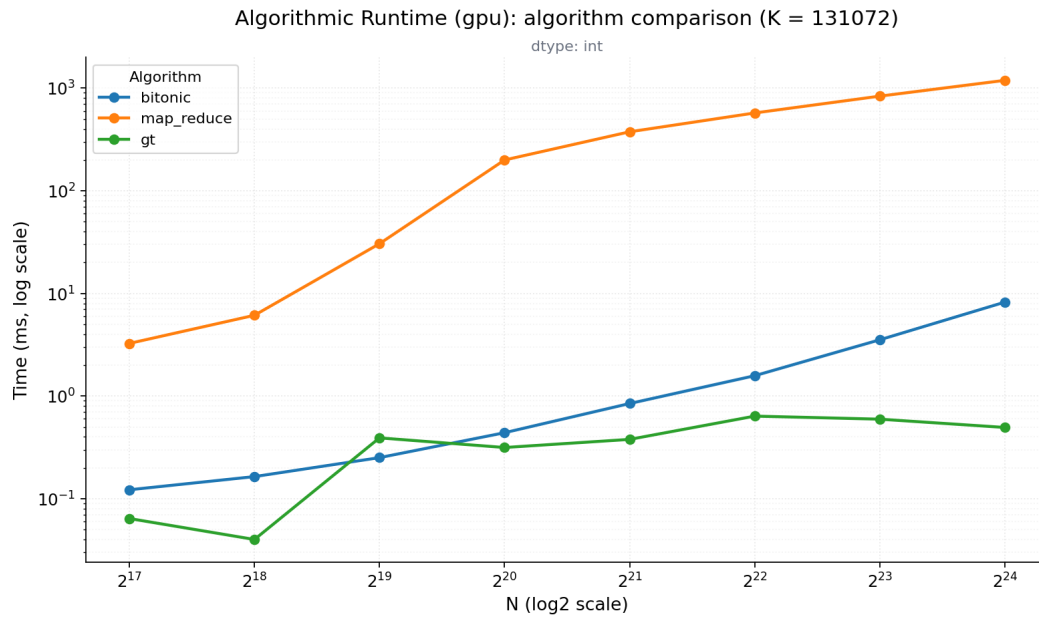


Figure 7.5: The three algorithms on the GPU for k equal to 131072.

The practical conclusion is that there is no universally preferable algorithm. The choice between a sorting network and heap-based selection depends on the ratio n/k and not on the architecture.

7.3 Comparison across architectures

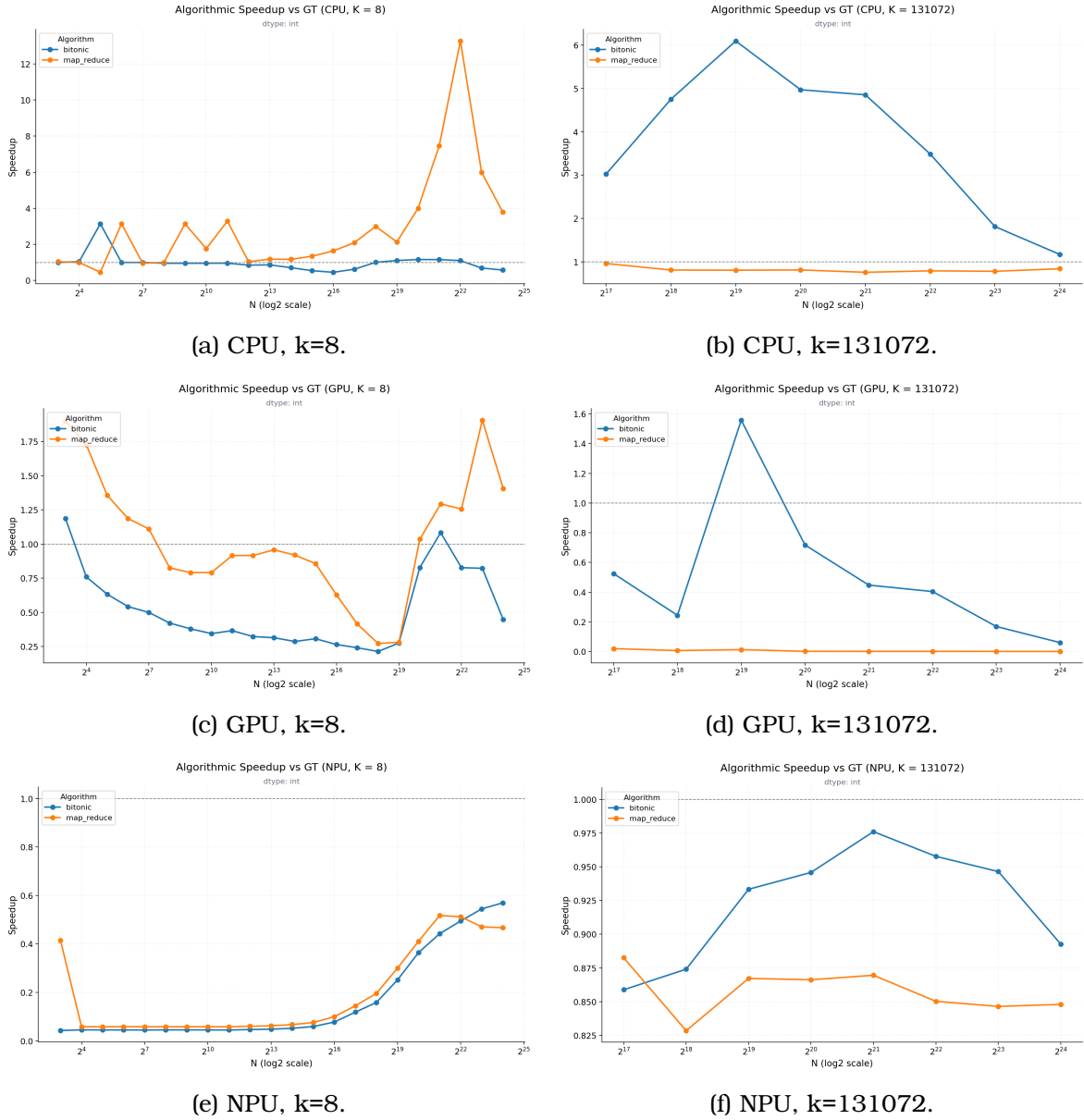


Figure 7.6: Speedup relative to the ground-truth implementation of the same architecture. Left column k equal to 8, right column k equal to 131072, one row per architecture. Values above unity mean that this work’s implementation is faster than the library.

Comparing each implementation with the reference of the same architecture gives the most honest picture, since it neutralises hardware differences.

On the CPU, map-reduce is faster than the library by a factor of three for small k . It should be recalled, however, that the limitation of section 4.5 applies: the library is single-threaded while this implementation uses eight threads, so this factor is not purely algorithmic.

The largest speedup in the entire figure appears elsewhere, however, and is a con-

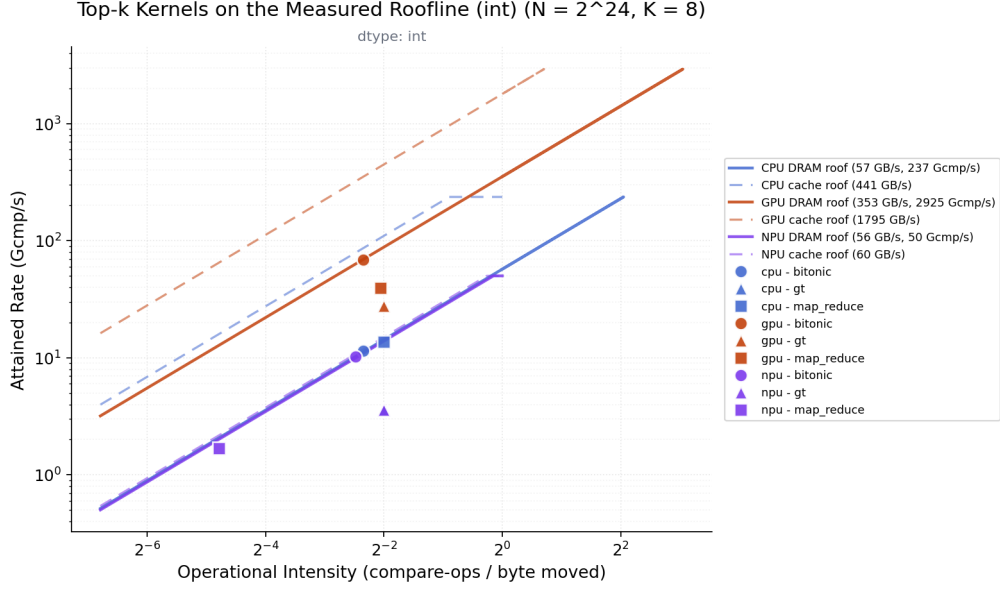
sequence of the reversal in section 7.2.2. For large k , where heap-based selection collapses, bitonic sort on the CPU reaches roughly six times faster than the library. The reason is that the library, too, relies on a heap of size k and suffers the same overhead, while the sorting network remains essentially unaffected. The advantage weakens as the input grows further and the network's data traffic starts to dominate, narrowing to just above unity at the largest input.

On the GPU the comparison is strict, since the library runs on the same device with the same parallelism. There, map-reduce marginally outperforms for k equal to 8, while for large k the library is faster by more than a thousand times. Radix selection has no point of collapse with respect to k , as explained in section 5.5.1.

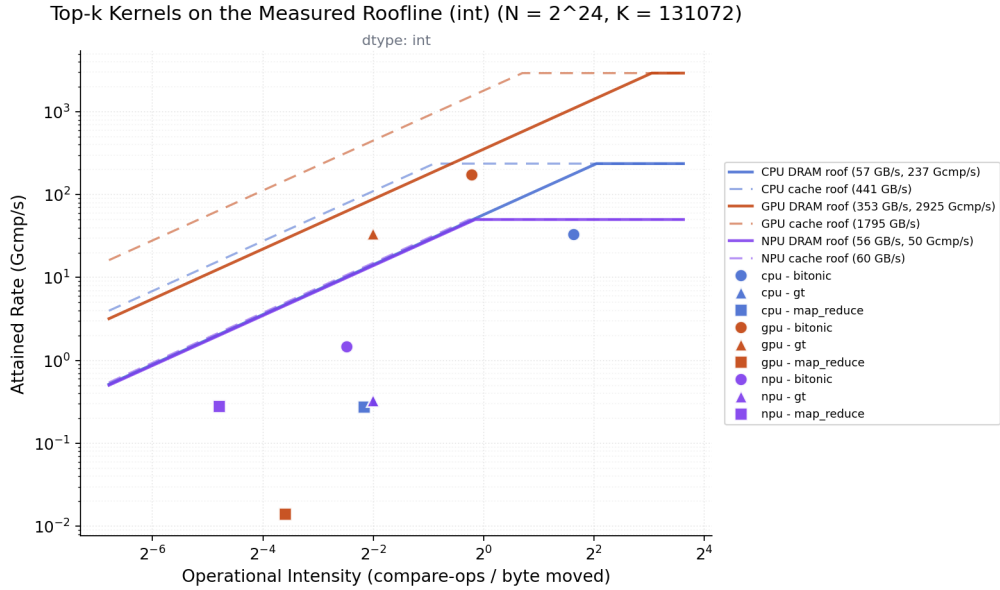
On the NPU, for the integer types to which the figure refers, no implementation surpasses the library in any parameter region. This result is expected rather than disappointing: as shown in section 6.6, the very processor that would run the library is in any case consumed by preparing the data for the device.

The qualifier “for the integer types” is not a hedge. For floating-point types, and especially for half precision, the picture reverses and the NPU does indeed surpass the library; the explanation is given in section 7.6, after the behaviour across types is first presented.

7.4 Roofline analysis



(a) k equal to 8.



(b) k equal to 131072.

Figure 7.7: Placement of the runs against the measured ceilings, for the largest input. Restricted to the maximum size, where the working set exceeds the cache and the main-memory ceiling is the relevant one.

The diagram places each run at its level of arithmetic intensity and achieved throughput. All implementations cluster in the left-hand region, i.e. at very low arithmetic intensity, and none approaches the horizontal part of the ceiling. No implementation of this work, on any architecture, is limited by compute capability. This finding agrees with an independent measurement on data-center accelerators, where the fastest known selection kernel utilises over ninety percent of the available band-

width against about half of compute capability, and where time scales with each device’s bandwidth [3].

Juxtaposing the two values of k adds two observations. In both cases bitonic sort on the GPU sits practically on the sloped section of its own ceiling, which visually confirms the finding of section 5.6: the implementation utilises the available bandwidth almost fully and is limited exclusively by the volume it is obliged to move.

The second observation concerns map-reduce on the same architecture. For k equal to 8 it sits near the other points, while for the large value it collapses by three orders of magnitude and moves away from both ceilings equally. This position, far from both the sloped and the horizontal section, is the signature of a limitation that is neither memory nor compute, and corresponds to the collapse of parallelism described in section 5.3.1.

This finding is structural rather than incidental. Top- k selection performs essentially one comparison per element and reuses almost nothing, so its arithmetic intensity is low by definition.

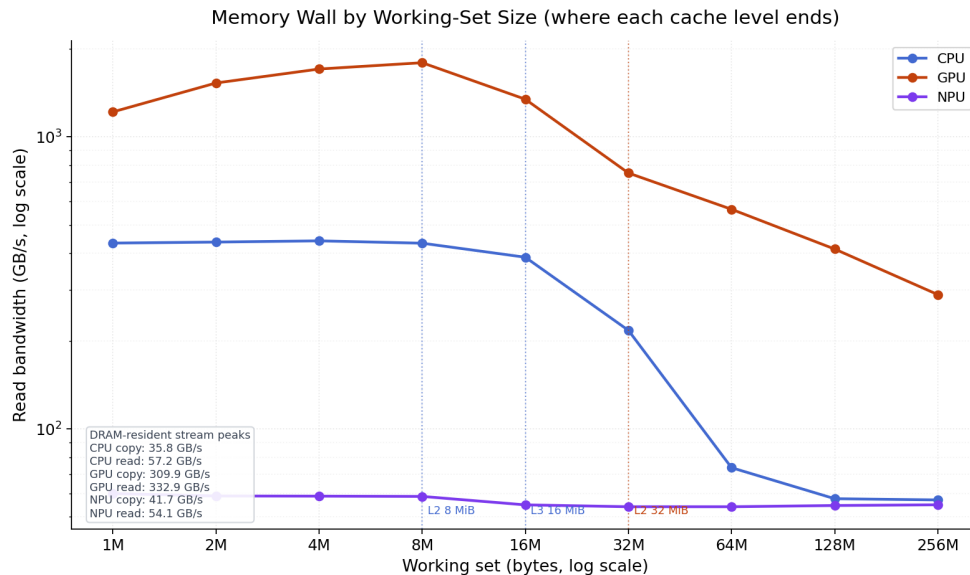


Figure 7.8: Measured read bandwidth as a function of working-set size, for the three architectures.

The ceiling against which each run is compared is not fixed but a function of its working set, per section 3.4. Figure 7.8 shows the measured ladder: the steps correspond to the levels of the memory hierarchy, and the drop toward the right-hand edge marks the point where the working set stops fitting in the last level.

7.4.1 Memory bandwidth utilisation

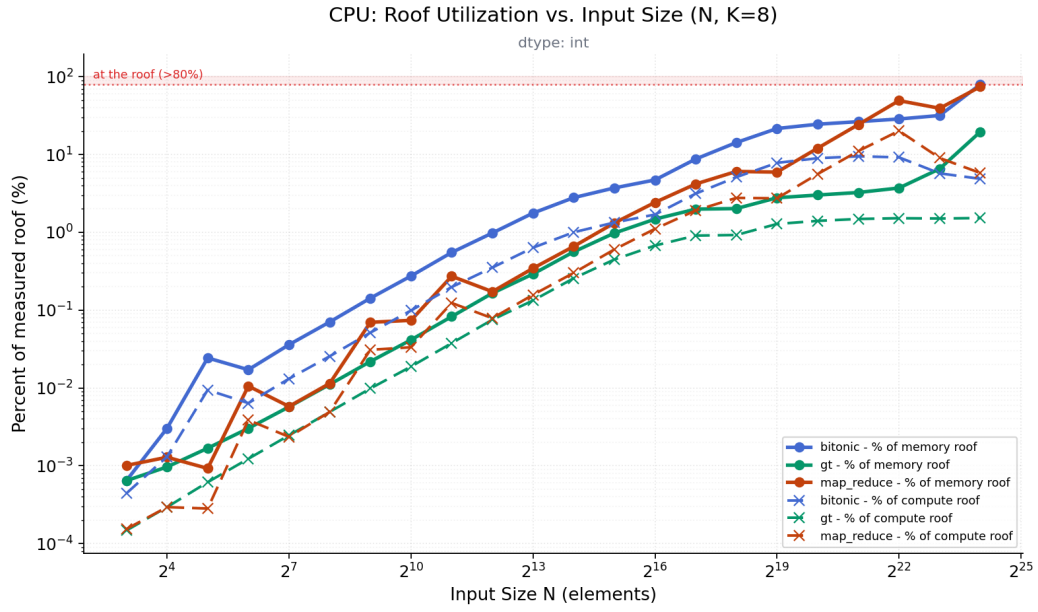
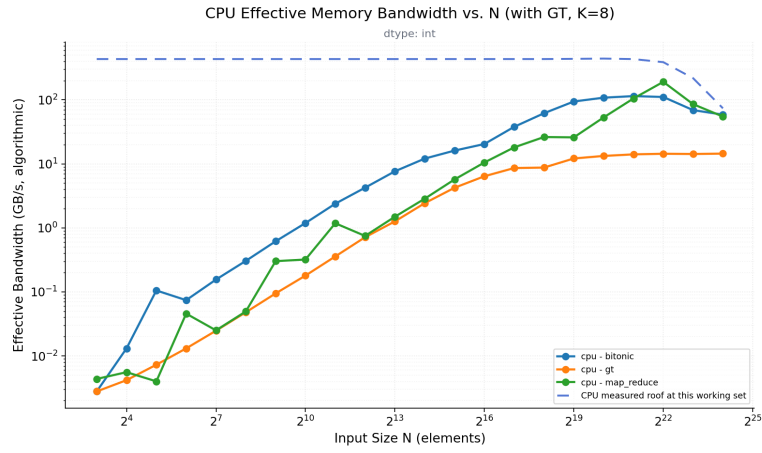
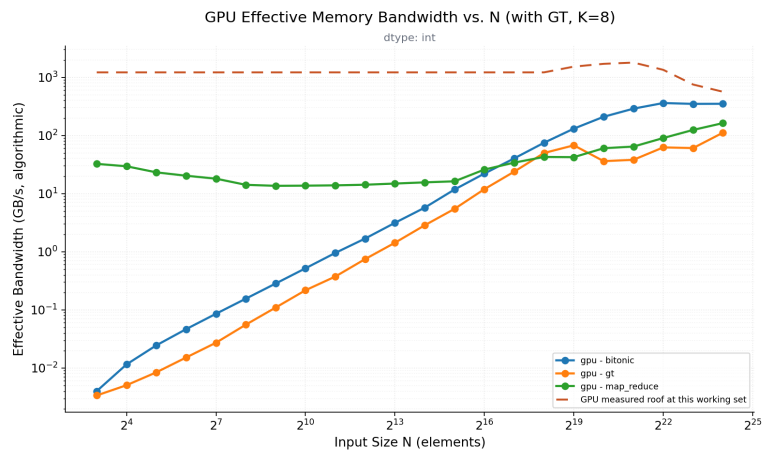


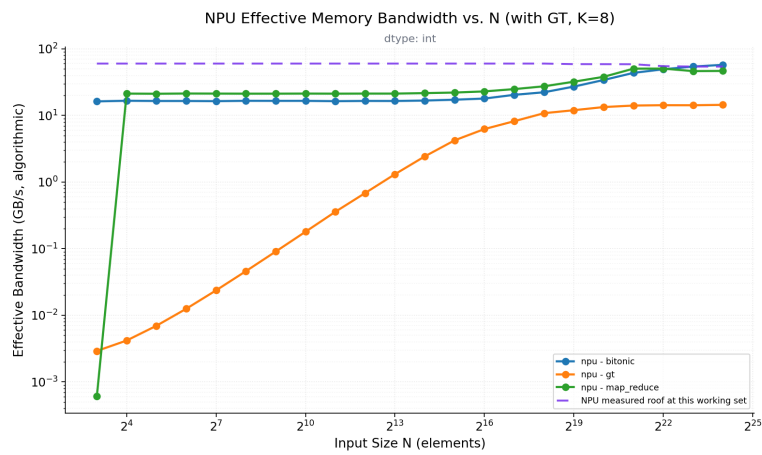
Figure 7.9: Utilisation of the two ceilings on the CPU, for k equal to 8.



(a) CPU.



(b) GPU.



(c) NPU.

Figure 7.10: Achieved bandwidth as a function of input size, one architecture per panel, with the measured ceiling drawn as a dashed line. The distance of each curve from the dashed line is the unexploited margin.

Utilisation of the memory ceiling increases with input size and stabilises once the working set exceeds the cache. At the largest input, bitonic sort on the CPU reaches

about four-fifths of the ceiling, noticeably ahead of the GPU's three-fifths, while utilisation of the compute ceiling remains below five percent on both.

The juxtaposition of bitonic sort with the library on the GPU deserves particular attention, because it shows that proximity to the ceiling does not equate to speed. Bitonic sort clearly utilises a larger fraction of the available bandwidth than the library, and is nonetheless slower, because it moves seven times the volume of data. Utilisation measures the quality of the implementation; total volume measures the quality of the algorithm.

7.5 Energy assessment

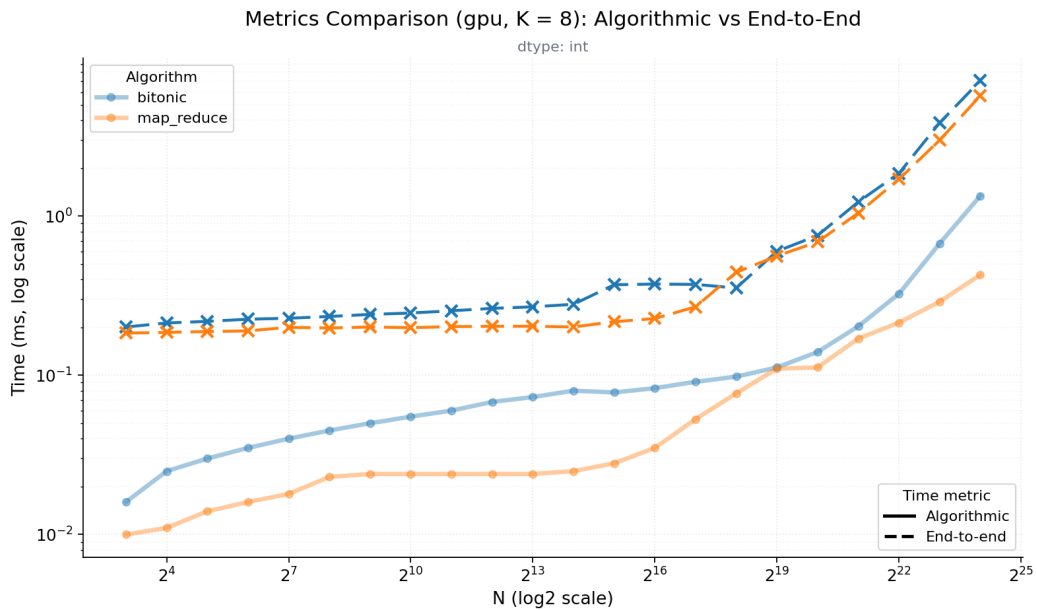


Figure 7.11: Algorithmic and end-to-end time on the GPU, for k equal to 8. The distance between the two curves of each algorithm is the cost of transfers and kernel launches.

Before presenting the energy measurements, figure 7.11 is useful, since it shows why the two measurement scopes diverge so sharply on the GPU. The end-to-end curve remains steady near three tenths of a millisecond up to about 2^{16} elements: throughout this range the kernel is essentially invisible and the time is spent entirely on transfers and launches. Even at the largest input, the end-to-end time remains roughly three times the algorithmic time.

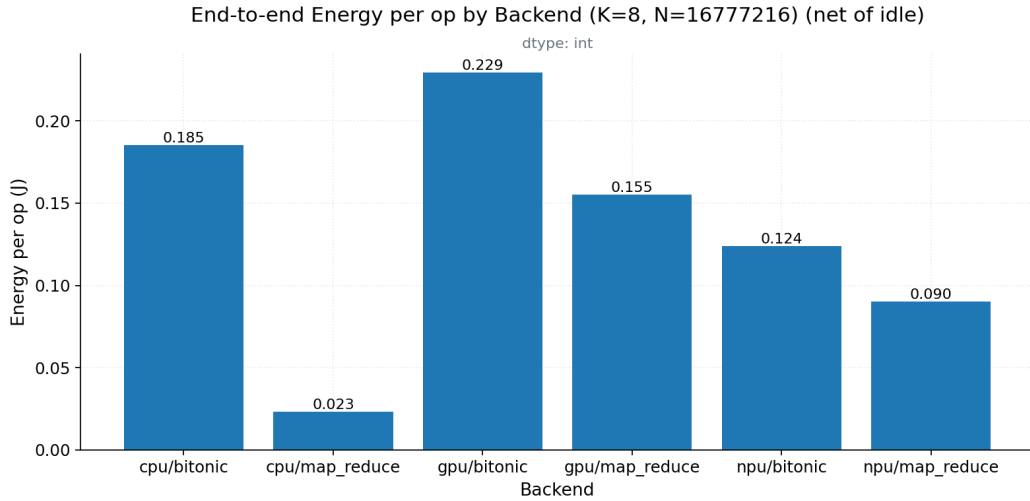


Figure 7.12: Net energy by architecture and algorithm, for k equal to 8.

The energy picture differs substantially from the time picture, and the difference is due to the symmetric measurement scope of section 3.3.1: each architecture is charged the total cost of solving the problem, including the host.

At the largest input and for small k, CPU map-reduce consumes about 19 mJ per run and is the most economical implementation. The corresponding implementations on the GPU consume from 108 mJ to 128 mJ, i.e. up to seven times more, despite being faster in the algorithmic scope. The most expensive implementation of all is bitonic sort on the GPU, while the corresponding CPU implementation costs 118 mJ.

The apparent contradiction is explained by section 5.4. The GPU's algorithmic scope includes only the kernel, while the end-to-end scope includes the transfers, which consume time and power on both sides of the bus. A device that is four times faster at the kernel but needs twice the kernel's own time just to receive the data, while simultaneously drawing board power, is not more economical.

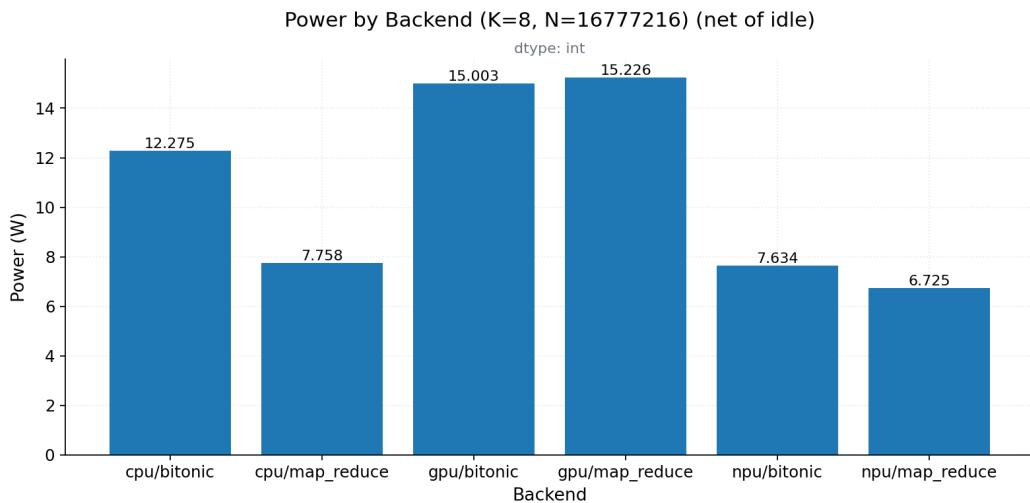


Figure 7.13: Net power by architecture and algorithm, for k equal to 8.

Juxtaposing figures 7.12 and 7.13 highlights a distinction that is easy to conflate, and one that particularly concerns the NPU. This device has the *lowest* power draw of all implementations, with map-reduce measuring 6.7 W against 7.8 W for the corresponding CPU implementation and 15.2 W for the GPU.

Nonetheless it consumes over four times the energy of CPU map-reduce, 0.080 J against 0.019 J. The reason is that energy is the product of power and time, and the NPU needs roughly eight times the time for the same task. The thirteen-percent advantage in power is wiped out by the eightfold disadvantage in duration.

This observation generalises: low power does not imply low energy. The principle that moving data costs more energy than the computation itself does [10] holds here as well; what is missing is the reuse that would amortise it. A low-power architecture pays off energetically only if it completes the task in comparable time, and for this particular problem that is not the case.

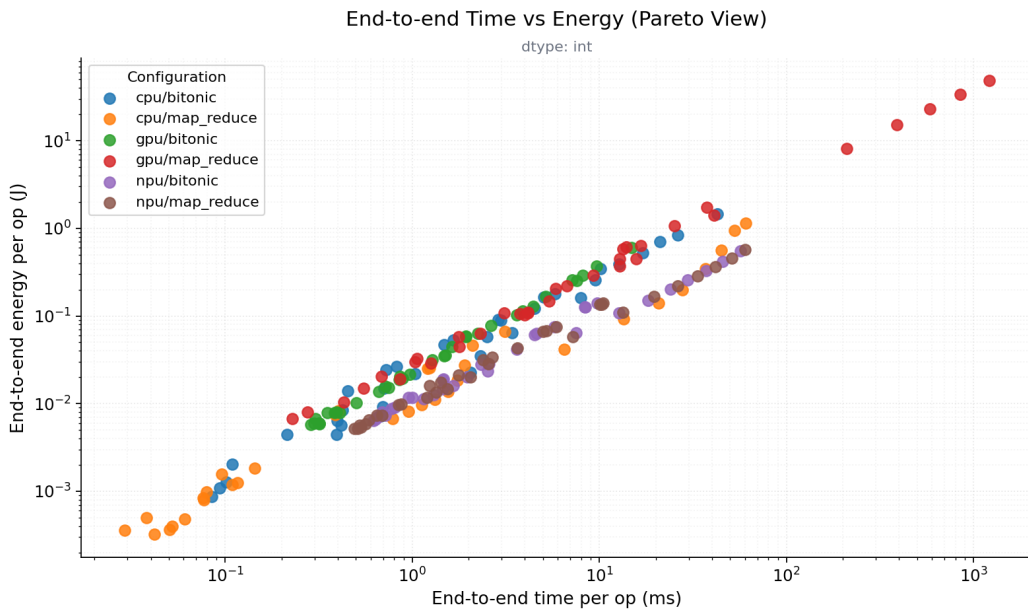


Figure 7.14: End-to-end time and total energy of each measured case. Both quantities are minimised, so the preferable cases lie toward the bottom-left corner.

Figure 7.14 places each measured case in a plane of two criteria that are minimised simultaneously. A case is said to be dominated by another when the latter is no worse in either quantity and is better in at least one; such a case is never worth choosing. The set of cases dominated by none other constitutes the Pareto front, i.e. exactly those choices in which improving one criterion requires giving up ground on the other.

The result is illuminating precisely because it is negative. The measurements form a narrow diagonal band, with a correlation coefficient between the logarithms of time and energy equal to 0.97. Of the 209 cases, only two belong to the front, and both are the same implementation, CPU map-reduce, at two neighbouring input sizes. That is, no substantial trade-off exists between speed and energy for this problem: the fastest configuration is also the most economical.

The reason follows from the two preceding figures. Power varies among implementations by barely a factor of 2.3, while time varies by four orders of magnitude. When one factor of the product is nearly constant, the product follows the other. The possibility of trading longer duration for lower power, which on other systems is a genuine design lever, does not exist here.

Finally, it is worth noting which cases make up the front: all of them run on the CPU. No configuration of the GPU or the NPU is Pareto-optimal at any point of the parameter space examined.

7.5.1 Energy-delay product

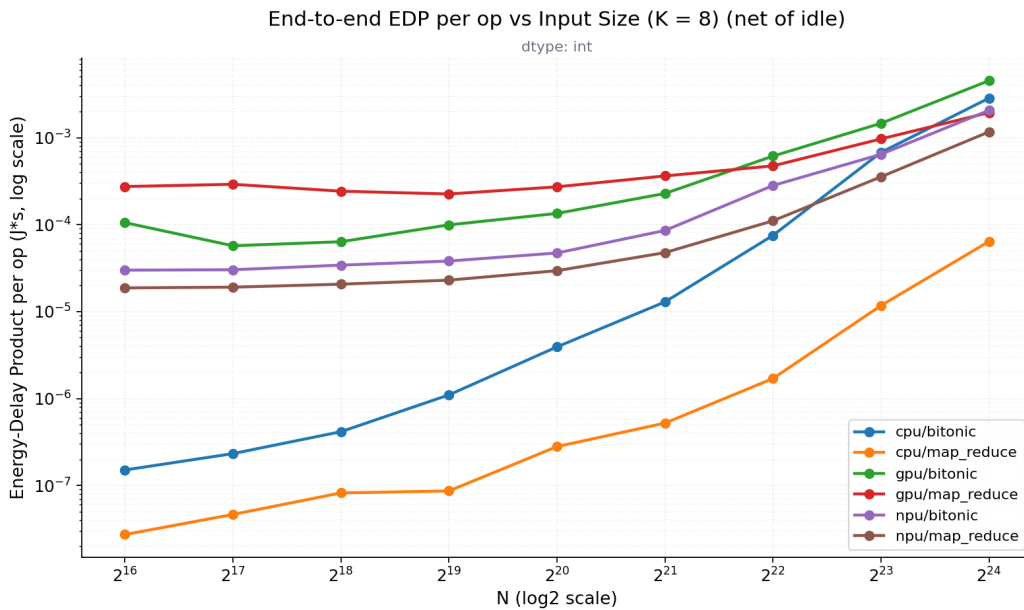


Figure 7.15: Energy-delay product as a function of input size.

The energy-delay product condenses the two criteria into one and rewards implementations that are simultaneously fast and economical. The resulting ranking is the clearest in the entire chapter. CPU map-reduce has the lowest product of any implementation, with the next-lowest, GPU map-reduce, trailing by a factor of twenty-eight, and the worst by a factor of forty-three.

CPU map-reduce is the lowest-product implementation across almost the entire range; the sole exception is the smallest size shown, where its net energy reading falls to zero, below the resolution of the counter at that duration, leaving CPU bitonic nominally lowest there instead. Elsewhere, the size of the gap to the next implementation is not monotonic: it fluctuates between roughly one and two and a half orders of magnitude, as CPU bitonic and CPU map-reduce trade the runner-up position depending on the exact input size. The ranking itself is never seriously in doubt; only the margin is noisy.

This result is the most counterintuitive of those presented here. For top-k selection with small k, offloading to an accelerator does not merely fail to pay off energetically; it is clearly worse than not offloading at all.

7.6 Effect of data type

Implementation	half	int	uint	float	double
CPU bitonic	4.60	8.01	7.93	8.01	14.03
CPU map-reduce	0.39	1.23	1.22	1.22	2.35
CPU reference	13.27	4.65	4.63	5.50	5.54
GPU bitonic	0.88	1.34	1.37	1.34	2.89
NPU bitonic	7.71	8.17	8.37	8.23	25.38

Table 7.2: Algorithmic time by data type, in ms, for 2^{24} elements and k equal to 8.

This work’s implementations track the width of the type, as expected of algorithms limited by memory. Half precision is consistently the fastest, with CPU bitonic sort completing in just over half the time of single precision, precisely because it moves half the bytes at the same rate of comparisons.

The reference implementation behaves oppositely: half precision is its slowest case. The difference reveals that the library is not limited by memory but by the cost of the scalar comparison, which for half precision includes a conversion. The ranking among types is therefore not a property of the hardware but of the implementation.

Double precision on the NPU stands out, with three times the time of any other type. The overhead is due entirely to the additional refinement pass of section 6.4.3, which is required because the 32-bit key is not sufficient to express 64-bit values.

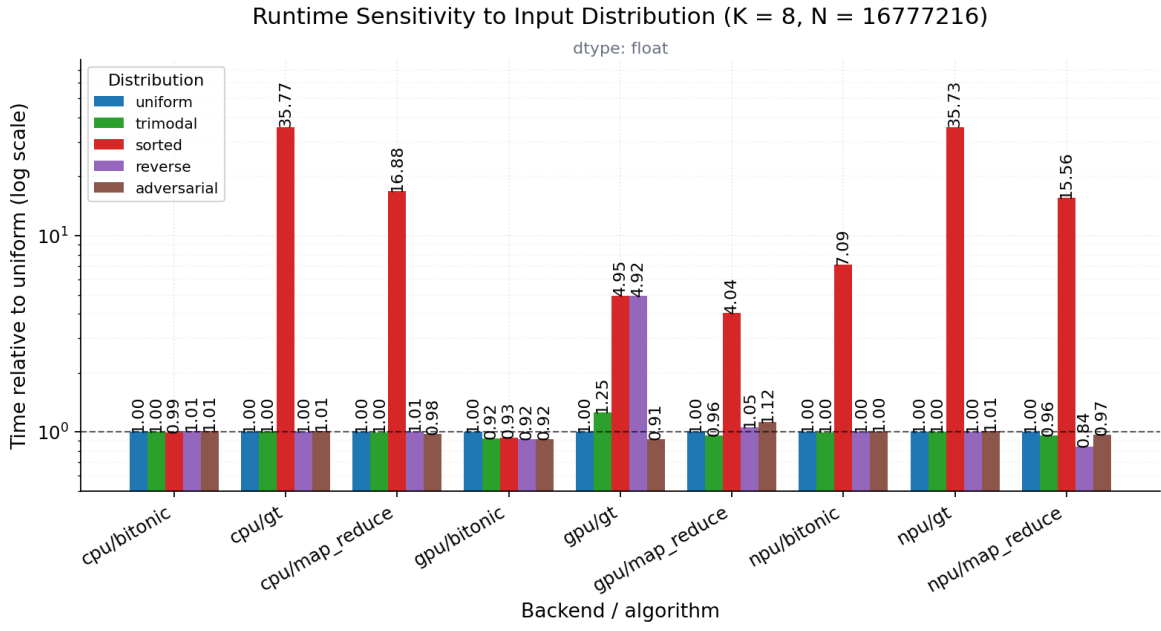
The combination of these two opposite behaviours produces the single point in the entire work where the NPU outperforms its own library. For half precision, bitonic sort on the NPU is faster than the library in twenty-nine of the measured cases, with the largest gap reaching a factor of two and a half. The phenomenon appears more weakly in the other two floating-point types, and never appears for the integer types.

The cause is not that the NPU handles half precision well, but that the library handles it poorly. The array’s kernel operates on 32-bit integers regardless of type, so the cost of the type shifts to host encoding, which for half precision is vectorised per section 6.4.2. The library, by contrast, performs scalar comparisons with an intermediate conversion. Offloading thus proves advantageous precisely where the established solution is weakest, not where the NPU is strongest.

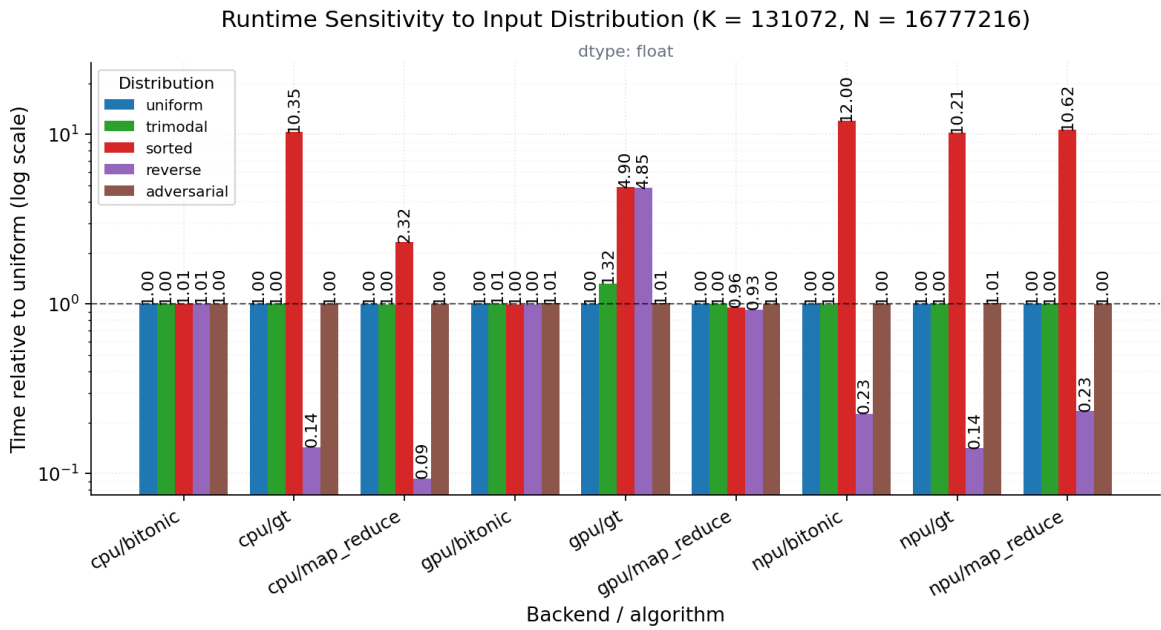
7.7 Sensitivity to the input distribution

Implementation	uniform	adversarial	sorted	reverse
CPU bitonic	42.98	43.20	43.16	43.05
GPU bitonic	8.08	8.24	8.09	8.13
CPU reference	73.97	73.91	765.13	10.60
NPU bitonic	56.69	56.56	678.15	12.78
GPU map-reduce	1190.89	1190.89	1138.59	1105.88

Table 7.3: Algorithmic time by input distribution, in ms, for single precision, 2^{24} elements, and k equal to 131072.



(a) k=8.



(b) k=131072.

Figure 7.16: Sensitivity of time to the input distribution, for single precision and 2^{24} elements. Each bar gives the time of one distribution relative to the time of the uniform distribution for the same implementation; unity means complete independence from the data.

Normalising against the uniform distribution allows all nine combinations to be compared on the same figure, even though their absolute times differ by orders of magnitude. An implementation unaffected by the data appears as five equal bars of height one, while a sensitive one appears with bars diverging in both directions.

The figure first confirms the expected: bitonic sort on the CPU and the GPU is

practically unaffected by the distribution at both values of k , with overall variation below a factor of 1.12. This property makes it a reliable witness: if its time were to change, the change would be attributable to the measurement rather than to the algorithm.

Heap-based implementations show the opposite extreme, and indeed regardless of who wrote the code. The reference library varies by a factor of seventy-two between ascending and descending input, but so does our own CPU map-reduce, by a factor of twenty-five, through the same mechanism described in section 4.3. Sensitivity to the data is therefore a property of the algorithm rather than of the implementation: whoever chooses a heap inherits it.

Two observations deserve particular attention. First, bitonic sort on the NPU is sensitive to the distribution, unlike the other two architectures, with a variation factor of fifty-three. This sensitivity does not originate in the array, which executes a fixed network, but in the heap-based merge performed by the host, which, as shown in section 6.4.1, dominates the time.

Second, juxtaposing the two values of k isolates the cause of the sensitivity. GPU map-reduce is sensitive for k equal to 8, where the sorted input slows it down by a factor of 4.3, but becomes nearly unaffected for k equal to 131072, where the same input changes it by only 4%. The same code, with the same data, changes behaviour because the limiting factor changes: in the second case it is not rejection that limits it but the lack of parallelism described in section 5.3.1. When the limiting factor is not the data, their distribution ceases to matter.

The measurements presented cover the entire parameter space of section 3.7 and agree with one another as to which factor limits each implementation. Their interpretation, the limits within which they hold, and the directions they open up are examined in chapter 8.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This thesis implemented two top-k selection algorithms on three architectures and measured them over the same parameter space, with a common harness and common correctness checking. The conclusions that follow do not repeat the measurements of Chapter 7, but rather what generalises from them beyond the specific hardware.

Top-k selection is a data-traffic problem. None of the nine configurations measured is bound by computational throughput. In the analysis of Section 7.4, every kernel is placed on the sloped segment of the roofline, that is, in the region where performance is determined by bandwidth rather than by the rate of comparisons. The practical consequence is that optimisation means reducing the volume of data moved rather than speeding up the computation. The thesis itself confirms this negatively: bitonic sort on the graphics processing unit clearly makes use of a larger fraction of its roofline than the reference library, and yet still loses, because it moves seven times as much data.

The appropriate algorithm is determined by the ratio n/k , not by the architecture. The inversion in Section 7.2.2 appears across all three architectures. The fastest implementation for k equal to 8 is the slowest for k equal to 131072, a slowdown by a factor of 2800, while the sorting network on the same device is penalised by only a factor of six. The cause is structural rather than implementation-specific: the heap only begins to filter once it is full, so the available parallelism is capped by the ratio n/k . The fact that independent work on different hardware arrives at the same distinction and addresses it with adaptive algorithm switching [15] reinforces the conclusion that this is a property of the algorithms.

The measurement boundary determines which architecture "wins". In the algorithmic scope the graphics processing unit is faster across the entire range of sizes; in the end-to-end scope the processor is faster across that same entire range. The ranking, that is, does not invert somewhere in the middle, but completely, and the only difference between the two answers is where the measurement boundary is

placed. This finding has methodological significance beyond the specific problem: any speed-up claim for offloading to an accelerator that reports only kernel time is not verifiable, because it conceals exactly the cost that determines whether offloading is worthwhile.

Low power does not imply low energy. The neural processing unit has the lowest power draw of any implementation, 6.7 W against 7.8 W for the processor and 15.2 W for the graphics processing unit, and at the same time consumes over four times more energy than the processor. The thirteen-percent advantage in power is wiped out by the eightfold disadvantage in duration. Moreover, as shown in Section 7.5, there is no substantial trade-off between time and energy for this problem: power varies among the implementations by only a factor of 2.3, while time varies by four orders of magnitude, so the product follows time. Of the 209 measured cases, only two belong to the Pareto front, and both run on the processor.

Overall, for top-k selection with small k, offloading to an accelerator does not merely fail to deliver a benefit; it is measurably worse than not offloading at all, with the energy-delay product trailing by a factor of between twenty-eight and seventy-eight relative to map-reduce on the processor. The conclusion does not concern the quality of the accelerators but the match between workload and architecture. The design principle of neural processing units, as stated in Section 2.3.3, is to maximise local reuse of data so as to amortise the energy cost of moving it. Top-k selection offers no reuse to exploit, so this design assumption simply does not hold. The measurement in Section 6.4.1, where 61% of the time is spent preparing and sending data and only 7% on waiting for the device, is the quantitative expression of this mismatch.

Finally, the sole exception deserves comment. For half-precision numbers, bitonic sort on the neural processing unit outperforms its ground truth in twenty-nine measured cases. The analysis in Section 7.6 shows, however, that this advantage arises because the library handles this particular type poorly, not because the device handles it well. This example is instructive methodologically: a speed-up over a baseline substantiates a claim only once it is established which of the two terms of the ratio caused it.

8.2 Limitations

The conclusions above hold within specific limits, which are stated here collectively.

The most significant asymmetry concerns the processor’s baseline. `std::partial_sort` is strictly single-threaded, whereas this thesis’s implementations use up to eight threads. Every ratio of times between them therefore conflates two causes, the difference in algorithm and the difference in core count, as already noted in Section 4.5. A thread-for-thread comparison would be stricter and would yield smaller speed-ups. This choice is justified by the question posed by the thesis, which concerns comparison with what the programmer already has available, but the reader should bear it in mind.

The measured data traffic of the reference implementations is a lower bound rather

than an actual measurement, since only the input and output are counted while the internal passes of the libraries are not observable. Every roofline utilisation percentage reported for them in Section 7.4.1 is consequently an underestimate and is not directly comparable with the corresponding percentages of this thesis’s two implementations.

For the neural processing unit there is no on-device baseline, because no top-k selection library exists for that architecture. The ground truth runs on the host, so the relevant comparisons answer the question "is offloading worthwhile" rather than the question "how good is the implementation relative to the best possible one on the device".

The phase-breakdown table in Section 6.4.1 comes from a manual run with separate instrumentation, not from the sweep that produces the rest of the results. It is therefore not reproduced by the procedure in Appendix B and should be read as an indicative breakdown of time rather than as a measurement of equivalent rigour to the others.

Finally, all measurements come from a single machine, with five data types, four values of k , and input sizes up to 2^{24} elements. The qualitative conclusions, in particular the dependence on the ratio n/k and the role of the measurement boundary, are expected to carry over to comparable systems; the specific numerical factors, however, should not be assumed to transfer.

8.3 Future Work

The clearest direction follows directly from the limiting factor identified for the neural processing unit. Since most of the time is spent preparing and sending data per call, a workload that executes many top-k queries at once would amortise exactly this cost. Batched selection, moreover, arises naturally in the applications mentioned in Section 2.1, where selection is repeated at every step or for every query. This extension would determine whether the unit falls behind because of its architecture or because of the way it is fed.

In the same direction, reducing the host’s involvement in pipelining, with the device accessing data directly instead of explicit copying per chunk, would attack the same cost from the other side. This possibility depends on the tools and interfaces exposed by the platform and is the most technically demanding of the open questions.

A second direction is adaptive algorithm selection. Given that the inversion in Section 7.2.2 is systematic and predictable from the ratio n/k , an implementation that dynamically chooses between the heap and the sorting network would cover the entire parameter range without a point of collapse. The transition point can be identified from this thesis’s own measurements.

A third direction is extending the parameter space toward points this thesis touched on but did not exhaust: values of k larger than those examined, where selection approaches sorting, and inputs that do not fit in a single device’s memory, where managing the stream becomes part of the algorithm rather than of the infrastructure.

Finally, the measurement harness developed here is independent of the specific problem. The separation of the two measurement scopes, the recording of energy with common semantics across three architectures, and the placement of kernels against a measured roofline are tools that can be applied to any workload being considered for offloading. Reusing them on a problem with substantial data reuse would provide the natural counterpart to the results presented here.

Appendix A

Acronyms

AIE	AI Engine
API	Application Programming Interface
AVX	Advanced Vector Extensions
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
D2H	Device to Host
DRAM	Dynamic Random Access Memory
EDP	Energy Delay Product
FIFO	First In First Out
GPU	Graphics Processing Unit
H2D	Host to Device
MLIR	Multi-Level Intermediate Representation
NPU	Neural Processing Unit
RAPL	Running Average Power Limit
SIMD	Single Instruction Multiple Data
SIMT	Single Instruction Multiple Threads
SM	Streaming Multiprocessor
XRT	Xilinx Runtime

Appendix B

Reproducing the Measurements

This appendix describes the steps required to reproduce the thesis’s measurements, from installing the tools to generating the plots. The commands are given for the system described in Section 3.1 and are executed from the root of the repository.

B.1 Software Requirements

All three implementations require a C++ compiler that supports the C++20 standard, together with CMake version 3.18 or newer. Vector instructions are not a compilation requirement: the choice between AVX-512, AVX2 and scalar execution is made at run time, based on the capabilities reported by the processor. The thesis’s measurements were obtained on a processor with AVX-512, so on a machine without it the CPU follows a different code path and its results are not directly comparable. Beyond these, each architecture has its own requirements.

The GPU requires the CUDA toolkit with the `nvcc` compiler and the CUB library, which is distributed together with it. Power measurements additionally require the NVML library, which ships with the driver.

The NPU requires the XRT runtime stack together with the XDNA device plugin, and, for compiling the kernels, MLIR-AIE within a Python environment. This environment must reside at the root of the repository under the name `npu_venv`, since both kernel compilation and the automated execution of the measurements look for it there. The user must belong to the groups that have access to the device, and the locked-memory limit must be sufficient; otherwise the device is not visible to the runtime stack. The `XILINX_XRT` variable must point to the installation location of XRT.

Energy measurement reads the RAPL counters through `/sys/class/powercap`, which requires elevated privileges. In a single run, without these the measurement is explicitly reported as unavailable rather than reporting zero values, as described in Section 3.3.3; the corresponding sweep, by contrast, does not even start and terminates with an error, so that no empty dataset is produced. Finally, the analysis infrastructure requires Python with the libraries declared in the repository; the corresponding environment is created automatically the first time it is run.

B.2 Compilation

Compiling all implementations is done with a single command:

```
make build-all
```

Each architecture can also be compiled separately, using the `build-cpu`, `build-gpu` and `build-npu` targets, which is useful on a system that does not have all three units. The resulting executables are placed in the `build` directory, two per architecture: one for the top-k measurements and one for the roofline measurements of Section 3.4.

Compilation for the NPU additionally produces the hardware configuration files. Because their absence leads to the corresponding measurements being silently skipped, their presence is checked explicitly:

```
make verify-artifacts
```

The check covers the two configuration files and the three top-k measurement executables, so it presupposes a full compilation of all three implementations; after compiling only a single architecture it fails by default.

B.3 Running the Experiments

The recommended way to reproduce the results is not to run the individual steps, but a single command that executes the entire procedure:

```
make benchmark GPU_W=50
```

The reason this is the preferred route is not convenience, but consistency. The order of the steps matters: if the system conditions are pinned after the measurements have started, or if a sweep is run with executables from an earlier compilation, the results are not comparable to one another. The command enforces the correct order and leaves no room for a step to be skipped.

The sequence that runs consists of six stages. First, the previous results and old executables are deleted, so that nothing survives from an earlier run. Next, all three implementations are compiled and the presence of the resulting files is checked, using the check from Section B.2. In the third stage, the system conditions are pinned and read access to the energy counters is granted. In the fourth, the four sweeps are run in sequence; in the fifth, the plots for each of them are generated; and in the sixth, the original system conditions are restored.

Two preconditions are checked before anything else: the availability of `sudo` and the existence of the `npu_venv` environment from Section B.1. If either is missing, the command terminates immediately, without altering the system or the existing results.

Three properties of the command are worth noting. Administrator privileges are requested only once, at the beginning, and remain active for the entire duration, so the run requires no further intervention afterwards. The conditions are restored in every case, even if the run is interrupted by the user or terminates with an error, so that the system is not left pinned. Finally, the failure of one sweep does not cancel

the rest: it is logged as a warning and the procedure continues, so that a partial run nonetheless produces whatever data it was possible to measure.

The GPU_W argument sets the power limit of the GPU, which, according to Section 3.8, must remain constant throughout the sweep. It should also be stressed that the command deletes the existing results as its first step, so any previous measurements that need to be kept must first be copied elsewhere.

B.3.1 Running Individual Steps

For a partial or repeated run, each stage is also available separately. The system conditions are controlled with:

```
make pin ARGS=50      # governor, boost off, GPU cap 50 W
make pin-show         # report the current state
make unpin            # restore governor, boost, GPU cap
```

The measurements are divided into four independent sweeps, each of which writes its raw results to a separate file:

```
make measure-cases    # timing over the parameter space
make measure-energy    # energy, requires RAPL access
make measure-dists    # input-distribution sensitivity
make measure-roofline # bandwidth, compare-exchange roofs
```

Each target accepts additional arguments through the ARGS variable, which are forwarded to the sweep driver and allow the parameter space to be restricted to a subset of it. This is useful for quickly checking the installation, where a full sweep would be excessively time-consuming. When the stages are run separately, responsibility for the correct order and for the stability of the conditions between sweeps rests with the user.

B.3.2 Execution Parameters

Table B.1 gives the full set of parameters accepted by each executable, their permitted values, their defaults, and the values with which the thesis's sweeps were run.

The defaults are those of the executable and in several cases do not match the values used in the sweeps. The value range extends up to 10^9 instead of the default, and the reason is substantive: with a range of only a few thousand values and an input of millions of elements, value repetitions would be so dense that the problem would turn into a choice among ties, which would artificially favour algorithms that reject quickly.

This range applies to four of the five types. For half, the executable clamps the bounds to ± 65504 , the largest finite value it can represent, so sampling actually takes place over $[0, 65504]$. For this type, ties are not avoided, both because of the narrower range and because above 2048 half does not represent every integer. This is an inherent property of the type rather than a methodological choice, but it must be taken into account when interpreting its results.

The seed is not fixed, but is generated deterministically from the q and k of each case, so that two different cases never receive the same input while each case remains reproducible.

The values in the fourth column refer to the main sweep. The two special sweeps restrict certain parameters: the distribution sweep examines q from 20 to 24, only the float and half types, and the k extremes 8 and 131072, with two seeds per case; the energy sweep examines q from 16 to 24 with three repetitions and without correctness checking, so that the check is not counted toward the consumption.

Parameter	Values	Default	In the sweeps	Meaning
q	integer 0-63	—	3-24	Input size $n = 2^q$; the only mandatory parameter
k	positive integer	n	8, 256, 4096, 131072	Number of requested elements; capped at n
mode	min max	max	max	Selects the smallest or largest elements
dtype	int uint float double half	int	all five	Element type
algo	bitonic map_reduce gt	bitonic	all three	Algorithm to run
run	full trunc both	trunc	trunc	Variant of the bitonic network
seed	non-negative integer	42	generated from q, k	Seed of the random-number generator
dist	uniform normal trimodal sorted reverse adversarial	uniform	uniform; five in the distribution sweep	Input distribution
min / max	integers	0 / 1000	0 / 10^9	Value range; requires $\max \geq \min$; clamped to the range of the type
threads	positive integer	0 (automatic)	8	Number of threads; ignored by the NPU
verify	true false	false	true	Checks correctness against the ground truth
debug	true false	true	false	Verbose diagnostic output

Table B.1: The execution parameters and their defaults.

B.4 Generating the Plots

The analysis infrastructure reads the raw results of the sweeps, converts them into tables in CSV format, and generates the plots:

```
make plot ARGS="--plot all"
```

The `-plot` argument selects which plots will be produced, while `-input` specifies which sweep will be read; the default is the timing sweep. The energy, distribution and roofline plots read their own files, so it is usually preferable to use the combined targets, which run the measurement and the corresponding plot generation together:

```
make benchmark-cases
make benchmark-energy
make benchmark-dists
make benchmark-roofline
```

The resulting files are organised by data type and by measurement category, together with the tables from which they were derived. Plots that concern the machine itself, such as the bandwidth scale of Section 3.4, are placed in a separate directory, since they do not depend on the data type.

Two options affect the interpretation of the plots and are worth mentioning. `-energy-metric` selects between gross energy, net energy, or both together, which is also the executable's default; the thesis's sweeps explicitly request the net figure, for the reasons given in Section 3.3.3. `-error-bars` selects the kind of dispersion band drawn around the values, among standard deviation, deciles, or no band at all, with standard deviation as the default. A band is drawn only where there are at least two samples per point, namely in the distribution sweep, which uses two seeds per case, and in the energy sweep, which repeats each case three times. In the automated run, the bands are disabled for all four sweeps, so the plots it produces do not include them.

Bibliography

- [1] D. E. Knuth, *The Art of Computer Programming, Volume 3: Sorting and Searching*, 2nd ed. Addison-Wesley, 1998.
- [2] N. Sismanis, N. Pitsianis, and X. Sun, “Parallel search of k-nearest neighbors with synchronous operations,” in *2012 IEEE Conference on High Performance Extreme Computing (HPEC)*, Waltham, MA, USA, 2012, pp. 1–6.
- [3] J. Zhang, A. Naruse, X. Li, and Y. Wang, “Parallel top-K algorithms on GPU: A comprehensive study and new methods,” in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC ’23)*, Denver, CO, USA, 2023.
- [4] K. E. Batcher, “Sorting networks and their applications,” in *Proceedings of the AFIPS Spring Joint Computer Conference*, 1968, pp. 307–314.
- [5] A. Shanbhag, H. Pirk, and S. Madden, “Efficient top-K query processing on massively parallel hardware,” in *Proceedings of the 2018 International Conference on Management of Data (SIGMOD)*, Houston, TX, USA, 2018, pp. 1557–1570.
- [6] D. A. Palmer and M. Florea, “Neural processing unit,” U.S. Patent US8655815B2, assigned to the University of California, San Diego, 2014, application US13/011,727, filed 21 January 2011; granted 18 February 2014.
- [7] H. Esmailzadeh, A. Sampson, L. Ceze, and D. Burger, “Neural acceleration for general-purpose approximate programs,” in *45th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2012, pp. 449–460.
- [8] T. Chen, Z. Du, N. Sun, J. Wang, C. Wu, Y. Chen, and O. Temam, “DianNao: A small-footprint high-throughput accelerator for ubiquitous machine-learning,” in *19th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2014, pp. 269–284.
- [9] N. P. Jouppi, C. Young, N. Patil, D. Patterson *et al.*, “In-datacenter performance analysis of a tensor processing unit,” in *44th International Symposium on Computer Architecture (ISCA)*, 2017, pp. 1–12.
- [10] Y.-H. Chen, J. Emer, and V. Sze, “Eyeriss: A spatial architecture for energy-efficient dataflow for convolutional neural networks,” in *43rd International Symposium on Computer Architecture (ISCA)*, 2016, pp. 367–379.
- [11] AMD, “Versal adaptive SoC: The industry’s first adaptive compute acceleration platform,” AMD, White Paper, 2021, <https://www.amd.com/>

content/dam/amd/en/documents/products/adaptive-socs-and-fpgas/
versal/adaptive-soc-paper.pdf.

- [12] A. Rico, S. Pareek, J. Cabezas, J. Noguera *et al.*, “AMD XDNA NPU in Ryzen AI processors,” *IEEE Micro*, vol. 44, no. 6, 2024.
- [13] AMD/Xilinx, “MLIR-AIE: An MLIR-based toolchain for AMD AI engine-enabled devices,” <https://github.com/Xilinx/mlir-aie>, 2024.
- [14] NVIDIA, “CCCL: CUDA core compute libraries,” <https://github.com/NVIDIA/cccl>, 2024.
- [15] C. Lin, M. Yang, C. Wang, and C. Huang, “Adaptive top-K selection: Eliminating performance cliffs across all K values on AMD GPUs,” AMD ROCm Blogs, Feb. 2026, <https://rocm.blogs.amd.com/software-tools-optimization/adaptive-topk/README.html>.
- [16] Y. Samaga, V. Yerram, S. R. Babbula, P. Jain, and P. Netrapalli, “A faster generalized two-stage approximate top-K,” *Transactions on Machine Learning Research*, 2026, <https://openreview.net/forum?id=izqZ1Crpjz>.
- [17] AMD, “Bitonic SIMD sorting on AI engine for float datatypes,” Vitis Tutorials: AI Engine Development, 2024.1, 2024, <https://docs.amd.com/r/2024.1-English/Vitis-Tutorials-AI-Engine-Development/Bitonic-SIMD-Sorting-on-AI-Engine-for-float-Datatypes>.
- [18] S. Williams, A. Waterman, and D. Patterson, “Roofline: An insightful visual performance model for multicore architectures,” *Communications of the ACM*, vol. 52, no. 4, pp. 65–76, 2009.
- [19] Intel, “Intel intrinsics guide,” <https://www.intel.com/content/www/us/en/docs/intrinsics-guide/index.html>, 2024.
- [20] AMD, “IRON: A close-to-metal programming framework for AMD NPUs,” <https://github.com/amd/IRON>, 2024.
- [21] —, “AI engine kernel and graph programming guide (UG1079): Casting and datatype conversion,” 2024, <https://docs.amd.com/r/en-US/ug1079-ai-engine-kernel-coding/>.